Chapter 12

Speed, Time and Distance

The concepts of time distance are most important in the terms of competitive exams. The basic concept of time and distance is used in solving the question based on motion in a straight line. The applications of time & distance are used to solve the problems related to trains and races.

The relation between time, distance and speed is

Distance = Time × Speed

i.e, $D = T \times S \Rightarrow \text{Time (T)} = \dfrac{\text{Distance (D)}}{\text{Speed (S)}} \Rightarrow \text{Speed (S)} = \dfrac{\text{Distance (D)}}{\text{Time (T)}}$

Example: A car covers 200 km in 4 hours, then find the speed of the car.

Sol. We know that, $\Rightarrow \text{Speed (S)} = \dfrac{\text{Distance (D)}}{\text{Time (T)}}$

Required speed $= \dfrac{200}{4} = 50$ km/h

Conversion of Units: (i) When we convert km/h into m/s, we multiply the speed by $\dfrac{5}{18}$. i.e, $1\text{km/h} = \dfrac{5}{18}$ m/s.

(ii) When we convert m/s into km/h, we multiply the speed by $\dfrac{18}{5}$. i.e, $1\text{m/s} = \dfrac{18}{5}$ km/h

Example: Convert 72 km/h into m/s.

Sol. We know that, 72 km/h $= \left(72 \times \dfrac{5}{18}\right)$ m/s $= 4 \times 5 = 20$ m/s

Concept 1:

Average speed: A certain distance is covered at 'x' km/h and the same distance is covered at 'y' km/h then the average speed during the whole journey.

$$\text{Average speed} = \frac{2xy}{x+y} \text{ km/h}$$

Example: Rohit covers a certain distance by car driving at speed of 40 km/h and he returns back to the starting point riding on a scooter with a speed of 10 km/hr. Find the average speed of the whole journey?

Sol. Average speed $= \dfrac{2 \times 40 \times 10}{40 + 10} = \dfrac{2 \times 400}{50} = 16$ km/hr

Concept 2: A person covers a distance in T hours and the first half at S_1 km/h and the second half at S_2 km/h, then the total distance covered by the person.

$$D = \frac{2 \times T \times S_1 \times S_2}{S_1 + S_2}$$

Example: A car covers a distance in 10 hrs, the first half at 40 km/h and the second half at 20 km/h. Find the distance travelled by car?

Sol. Distance $= \dfrac{2 \times 10 \times 40 \times 20}{40 + 20}$ km/h $= \dfrac{2 \times 10 \times 40 \times 20}{60} = 266.67$ km

Concept 3: If two persons P and Q start at the same time in opposite directions from two points and after passing each they complete their journeys in 'a' and 'b' hrs respectively then

$$\frac{\text{P's speed}}{\text{Q's speed}} = \frac{\sqrt{b}}{\sqrt{a}}$$

Example: Shivam sets out to cycle from Delhi to Ghaziabad and at the same time Hemant starts from Ghaziabad to Delhi, After passing each other they complete their journeys in 4 and 16 hours respectively. At what rate does Hemant cycle if Shivam cycle at 18 km per hour?

Sol. $\frac{\text{Shivam's speed}}{\text{Hemant's speed}} = \frac{\sqrt{16}}{\sqrt{4}} \Rightarrow \frac{18}{\text{Hemant's speed}} = \frac{4}{2} \Rightarrow$ Hemant's speed $= \frac{18}{2} = 9 \text{ km/h}$

Concept 4: If a man travelled a certain distance by bus at a rate of x km/h and walked back at the rate of 'y' km/h. If the whole journey took 't' hours, then the distance he travelled is $\left(\frac{xy}{x+y}\right)t$ km.

Example: A man travelled a certain distance by train at a rate of 15 km/h and walked back at the rate of 12 km/h. The whole journey took 9 hours. Find the distance he travelled?

Sol. Required distance $= \left(\frac{xy}{x+y}\right)t = \left(\frac{15\times12}{15+12}\right)9 = \frac{15\times12\times9}{27} = 60$ km

Concept 5: If a person changes his speed to $\frac{a}{b}$ of its usual speed and late by T minutes, then the usual time taken by him is $\frac{T}{\frac{b}{a}-1}\left[\frac{a}{b}<1\right]$ and when $\left[\frac{a}{b}>1\right]$ then the usual time taken by him is $\frac{T}{1-\frac{b}{a}}$.

Example: Walking $\frac{4}{5}$ of his usual speed, a man is 16 minutes late. Find the usual time taken by him to cover that distance?

Sol: Usual time $= \frac{T}{\frac{b}{a}-1} = \frac{16}{\frac{5}{4}-1} = \frac{16\times4}{5-4} = \frac{16\times4}{1} = 64$ minutes

Concept 6: (i) If speed is constant, then distance is directly proportional to the time; $D \propto T$

(ii) If time is constant, then distance is directly proportional to the speed; $D \propto S$

(iii) If Distance is constant, then speed is inversely proportional to the time; $S \propto \frac{1}{T}$

Example A person covers a certain distance with a speed of 54 km/h in 15 min. If he wants to cover the same distance in 30 min, what should be his speed?

Sol. We know that, Distance = Speed × Time $= 54\times\frac{15}{60} = \frac{9}{10}\times15 = \frac{27}{2}$ km

Speed to cover $\frac{27}{2}$ km in 30 min $= \frac{\frac{27}{2}}{\frac{30}{60}} = \frac{27}{2}\times2 = 27$ km/h

Concept 7: (i) When a train passes a pole or any other object, the distance covered by train is equal to the length of the train.

(ii) If a train passes a bridge, platform etc, then distance travel by train is equal to the sum of the length of train and the stationary object through which the train is passing.

Example: A 100 m long train passes a platform of 200 m long. Find the distance covered by the train in passing the platform?

Sol. Required distance = length of train + length of platform = 100 + 200 = 300 m

Concept 8: (i) When two trains are moving is opposite directions, then their relative speed is equal to the sum of the speed of both trains.

(ii) When two trains are moving is same directions, then their relative speed is equal to the difference of the speed of both trains.

Example: Two trains are moving in the same direction with speed of 40 km/h and 50 km/h respectively. Find the relative speed?

Sol. Required relative speed = (50 – 40) km/h = 10 km/h

Concept 9: Two trains start at the same time from P and Q and proceed towards each other at the rate of x km/h and y km/h respectively. When they meet it is found that one train has travelled D km more than the other. The

Distance between P and Q is $\left(\frac{x+y}{x-y}\right)D$, Distance = $\frac{\text{Sum of speeds}}{\text{Difference of speed}} \times$ Difference in distance

Example: Two trains start at the same time from Kanpur and Delhi and proceed towards each other at the rate of 74 km/h and 47 km/h respectively. When they meet it is found that the train has travelled 13 km more than the other. Find the distance b/w Kanpur and Delhi?

Sol. Required Distance = $\frac{\text{Sum of speeds}}{\text{Difference of speeds}} \times \text{Difference in Distance} = \frac{(73+47)}{(73-47)} \times 13 = \frac{120}{26} \times 13 = 60$ km

Concept 10: When the speed of two trains are in the ratio x : y. They are moving in opposite directions on parallel tracks. The first train crosses a telegraph pole in 't_1'seconds where as the second train crosses the pole in 't_2' seconds. Time taken by the trains to cross each other completely is given by

Time taken = $\frac{t_1x+t_2y}{x+y}$ seconds.

Example: The speed of two trains are in the ratio 4 : 5. They are moving in opposite directions along the parallel tracks. If each takes 3 seconds to cross a pole. Find the time taken by the train to cross each other completely?

Sol. Time taken = $\left(\frac{t_1x+t_2y}{x+y}\right) = \frac{3\times4+3\times5}{4+5} = \frac{27}{9} = 3$ seconds

Types of Questions

1. A man covers a certain distance between his house and office on bike. When his speed is 30 km/h, he is late by 20 min. However, with a speed of 40 km/h, he reaches his office 10 min earlier. Find the distance between his house and office?

Sol. Let the distance be x km.

Time taken to cover x km at 30 km/h = $\frac{x}{30}$ h

Time taken to cover x km at 40 km/h = $\frac{x}{40}$ h

Difference between time taken = (20 + 10) m = $\frac{1}{2}$ h

$\therefore \frac{x}{30} - \frac{x}{40} = \frac{1}{2} \Rightarrow 40x - 30x = \frac{30\times40}{2}$

$x = \frac{30\times40}{20} = 60$ km

Hence, the required distance is 60 km.

2. A boy goes to school at a speed of 6 km/h and return to his house at a speed of 4 km/h. If he takes 5 hrs in all, what is the distance between his house and the school?

Sol. Let the required distance be x km.

Then time taken when he goes to school = $\frac{x}{6}$ h

Time taken when he goes to his house = $\frac{x}{4}$ h

Therefore, $\frac{x}{6} + \frac{x}{4} = 5$

$\Rightarrow \frac{2x+3x}{12} = 5$

$\therefore x = \frac{5}{5} \times 12 = 12$ km

3. The distance between two stations A and B is 450 km. A train starts from A and moves towards B at an average speed of 20 km/h. Another train starts from B, 20 minutes earlier than the train at A, and moves towards A at an average speed of 30 km/h. How far from A will the two trains meet?

Sol. Let the trains meet at a distance of x km from A.

$\frac{450-x}{30} - \frac{x}{20} = \frac{20}{60}$; $\frac{20\,(450-x)-30x}{20\times30} = \frac{1}{3}$

$9000 - 20x - 30x = \frac{20 \times 30}{3}$, $9000 - 50x = 200$

$8800 = 50x \Rightarrow x = \frac{8800}{50} = 176$ km

4. Two men P and Q start from a place a walking at 6 km and 8 km an hour respectively. How many km will they be apart at the end of 2 hours, if they walk in same direction?

Sol. Distance travelled by P in 2 hours = 2 × 6 = 12 km
Distance travelled by Q in 2 hours = 2 × 8 = 16 km
Distance between P and Q = (16 – 12) km = 4 km

5. Two runners cover the same distance at the rate of 10 km/h and 15 km/h respectively. Find the distance travelled when one takes 16 minutes longer than the other?

Sol. Let the distance be x km.

Time taken by the first runner = $\frac{x}{10}$ h

Time taken by the second runner = $\frac{x}{15}$ h

Now, $\frac{x}{10} - \frac{x}{15} = \frac{16}{60} \Rightarrow \frac{3x - 2x}{30} = \frac{16}{60}$

$x = \frac{16}{60} \times 30 = 8$ km

6. Without any stoppage a person travels a certain distance at an average speed 80 km/h and with stoppages he covers the same distance at an average speed of 60 km/h. How many minutes per hour does he stop?

Sol. Let the total distance be x km.

Time taken at the speed of 80 km/h = $\frac{x}{80}$ h

Time taken at the speed of 60 km/h = $\frac{x}{60}$ h

$\therefore$ he rested for $\left(\frac{x}{60} - \frac{x}{80}\right)$h $= \frac{20x}{60 \times 80} = \frac{x}{240}$ h

He rested per hour = $\frac{x}{240} \div \frac{x}{60} = \frac{1}{4}$ h = 15 minutes

7. A train leaves the station 1 hour before the scheduled time. The driver decreases its speed by 2 km/h. At the next station 60 km away, the train reached on time. Find the original speed of the train?

Sol. Let it takes x hours in second case

$\frac{60}{x-1} - \frac{60}{x} = 2$, $\frac{30}{x-1} - \frac{30}{x} = 1$

$30x - 30(x - 1) = x^2 - x \Rightarrow 30x - 30x + 30 = x^2 - x$
$x^2 - x - 30 = 0$
$x^2 - 6x + 5x - 30 = 0 \Rightarrow x(x - 6) + 5(x - 6) = 0$
$(x + 5)(x - 6) = 0 \Rightarrow x = -5, 6; \therefore x = 6$

Train takes 6 hours in second case
5 hours in original case

Original speed = $\frac{60}{5} = 12$ km/h

8. A thief is spotted by a policemen at a distance of 200 metres. When the policeman starts to chase, the theif also starts running. The speed of thief and policemen are 10 km/h and 14 km/h respectively. How far will have the thief run before he caught?

Sol. Relative Speed = (14 – 10) = 4 km/h

$\therefore$ The thief will be caught after = $\frac{200}{1000 \times 4} = \frac{1}{20}$ h

The distance covered by thief before he caught

$= 10 \times \frac{1}{20} = 0.5$ km = 500 m

9. The ratio between the speeds of Hemant and Nitish is 6 : 7. If Hemant takes 30 minutes more than Nitish to cover a distance. Find the actual time taken by Hemant and Nitish ?

Sol. Let the speed of Hemant and Nitish be x_1 km/h and x_2 km/h respectively.
and the time taken by Hemant and Nitish be t_1 and t_2 respectively.

Distance = $x_1 t_1 = x_2 t_2$, $\frac{x_1}{x_2} = \frac{t_2}{t_1}$ or $\frac{t_2}{t_1} = \frac{6}{7}$

$\frac{t_1 - t_2}{t_1} = \frac{7-6}{7}$, $\frac{t_1 - t_2}{t_1} = \frac{1}{7}$, $\frac{\frac{30}{60}}{t_1} = \frac{1}{7}$

$t_1 = \frac{7}{2}$ hours $t_2 = \frac{7}{2} \times \frac{6}{7} = 3$ hours

Actual time taken by Hemant = $\frac{7}{2}$ hours

Actual time taken by Nitish = 3 hours

10. A man tries to ascend a greased pole 101 metres high. He ascends 10 metres in first minute and slips down 1 metre in the alternate minute. If he continues to ascends in this way. How long does he takes to reach the top?

Sol. In every 2 minutes, he is able to ascend (10 – 1) = 9 metres
In 22 minutes, he ascends upto 9 × 11 = 99 metres

For remaining distance he takes = $\frac{2}{10} = \frac{1}{5}$ min = 12 seconds
Total time taken = 22 min 12 seconds.

11. A train running at 25 km/h takes 18 seconds to pass a platform. Next, it takes 12 seconds to pass a man walking at 5 km/h in the opposite direction. Find the length of the train and platform?

Sol. Speed of train relative to man = (25 + 5) = 30 km/h

$= 30\times\frac{5}{18}=\frac{25}{3}$ m/sec

Distance travelled in 12 seconds = $\frac{25}{3}\times 12 = 100$ m

Length of train = 100 m

Speed of train = 25 km/h = $25\times\frac{5}{18}=\frac{125}{18}$ m/sec

Distance travelled in 18 seconds = $\frac{125}{18}\times 18 = 125$ m

Length of train + length of platform = 125 m

Length of platform = (125 – 100) = 25 m

12. Two trains of length 200 m and 250 m respectively with different speeds passes a static pole in 8 sec and 14 sec respectively. In what time will they cross each other when they are moving in same direction.

Sol. Speed of first train = $\frac{200}{8}$ = 25 m/s

Speed of second train = $\frac{250}{14}=\frac{125}{7}$ m/s

Total distance to be travelled = 200 + 250 = 450 m

Relative speed when they are moving in the same direction

$= 25 - \frac{125}{7}$

Required time = $\frac{450\times 7}{50}$ = 63 sec

13. A train overtakes two persons who are walking in the same direction as the train is moving, at the rate of 2 km/hr and 4 km/hr and passes them completely in 9 and 10 seconds respectively. Find the speed and the length of the train?

Sol. Speed of two men are:

2 km/hr = $2\times\frac{5}{18}=\frac{5}{9}$ m/s and

4 km/hr = $4\times\frac{5}{18}=\frac{10}{9}$ m/s

Let the speed of the train be x m/s. Then relative speed are $\left(x-\frac{5}{9}\right)$ m/s and $\left(x-\frac{10}{9}\right)$ m/s

Now, length of train = Relative speed × Time taken to pass a man

$\therefore$ length of train = $\left(x-\frac{5}{9}\right)\times 9=\left(x-\frac{10}{9}\right)\times 10$

$\therefore\ x=\frac{100}{9}-\frac{45}{9}=\frac{55}{9}$ m/s

$\therefore$ speed of the train = $\frac{55}{9}\times\frac{18}{5}=22$ km/hr and

length of the train = $\left(x-\frac{5}{9}\right)9=\left(\frac{55}{9}-\frac{5}{9}\right)9=50$ m.

14. A train after travelling 50 km meets with an accident and then proceeds at $\frac{3}{4}$ of its former speed and arrives at its destination 35 minutes late. Had the accident occurred 24 km further, it would have reached the destination only 25 minutes late. Find the speed of the train and the distance which the train travels?

Sol. Let the speed of the train be x km/hr and the distance D km.

From the question we have,

$$\frac{50}{x}+\frac{(D-50)4}{3x}=\frac{D}{x}+\frac{35}{60}$$

or, $\frac{150+4D-200}{3x}=\frac{12D+7x}{12x}$

or, 4D – 7x = 200 ... (i) and

$$\frac{74}{x}+\frac{(D-74)4}{3x}=\frac{D}{x}+\frac{25}{60}$$

or, $\frac{222+4D-296}{3x}=\frac{12D+5x}{12x}$

4D – 5x = 296 ... (ii)

Now, subtracting equation (i) from the equation (ii), we have

2x = 96

$\therefore$ x = 48 km/hr

$\therefore$ Speed of the train = 48 km/hr

To find the distance, put the value of x in equation (ii)

4D – 5x = 296

or, 4D – 5 × 48 = 296

or, D = $\frac{536}{4}$ = 134 km

$\therefore$ Distance which the train travel = 134 km.

15. In a race of 2500 m, A beats B by 500 m and in a race of 2000 m, B beats C by 800 m. By what distance A gives start to C so that they will end up at same time in 3 km race. Also, find by what distance A will win over C in 1 km race?

Sol. Ratio of speeds of A : B = 2500 : 2000 = 5 : 4

Ratio of speeds of B : C = 2000 : 1200 = 5 : 3

Ratio of speeds of A : B : C = 25 : 20 : 12

In 3 km race A run 3000 m,

B run = $\frac{3000\times 20}{25}$ = 2400 m

C run = $\frac{3000\times 12}{25}$ = 1440 m

So to end up the race at same time A should gives C the start of 1560 m.

In 1 km race, A will win over C by 520 m.

Foundation

Questions

1. A man travels first 50 km at 25 kmph, next 40 km at 20 kmph and then 90 km at 15 kmph. His average speed for the whole journey (in kmph) is:
(a) 25 (b) 20
(c) 18 (d) 40

2. A man walks at the rate of 5 km/hr for 6 hours and at 4 km/hr for 12 hours. The average speed of the man (in km/hr) is:
(a) 4 (b) $4\frac{1}{2}$
(c) $4\frac{1}{3}$ (d) $4\frac{2}{3}$

3. If a person travels $10\frac{1}{5}$ km in 3 hours, then the distance covered by him in 5 hours will be :
(a) 18 km (b) 17 km
(c) 16 km (d) 15 km

4. If a train 110m long passes a telegraph pole in 3 seconds, then the time taken (in seconds) by it to cross a railway platform 165 m long, is:
(a) 3 (b) 4
(c) 5 (d) 7.5

5. A train 700 m long is running at the speed of 72 km/hr. If it crosses a tunnel in 1 minute, then the length of the tunnel (in metres) is :
(a) 700 (b) 600
(c) 550 (d) 500

6. If a 200 m long train crosses a platform of the same length as that of the train in 20 seconds, then the speed of the train is :
(a) 50 km/hr (b) 60 km/hr
(c) 72 km/hr (d) 80 km/hr

7. Two trains, each of length 125 metre, are running in parallel tracks in opposite directions. One train is running at a speed 65 km/hour and they cross each other in 6 seconds. The speed of the other train is:
(a) 75 km /hour (b) 85 km/ hour
(c) 95 km/ hour (d) 105 km/ hour

8. A man with $\frac{3}{5}$ of his usual speed reaches the destination $2\frac{1}{2}$ hours late. Find his usual time to reach the destination?
(a) 4 hours (b) 3 hours
(c) $3\frac{3}{4}$ hours (d) $4\frac{1}{2}$ hours

9. A train running at $\frac{7}{11}$ of its normal speed reached a place in 22 hours. How much time could be saved if the train would have run at its normal speed?
(a) 14 hours (b) 7 hours
(c) 8 hours (d) 16 hours

10. Walking at three-fourth of his usual speed, a man covers certain distance in 2 hours more than the time he takes to cover the distance at his usual speed. The time taken by him to cover the distance with his usual speed is:
(a) 4.5 hours (b) 5.5 hours
(c) 6 hours (d) 5 hours

11. A man goes from a place A to B at a speed of 12 km/hr and returns from B to A at a speed of 18 km/hr. The average speed for the whole journey is:
(a) $14\frac{2}{5}$ km/hr (b) 15 km/hr
(c) $15\frac{1}{2}$ km/hr (d) 16 km/hr

12. Two trains started at the same time, one from A to B and the other from B to A. If they arrived at B and A respectively 4 hours and 9 hours after they passed each other, the ratio of the speeds of the two trains was:
(a) 2 : 1 (b) 3 : 2
(c) 4 : 3 (d) 5 : 4

13. A starts from a place P to go to a place Q. At the same time B starts from Q to P. If after meeting each other A and B took 16 and 25 hours more respectively to reach their destinations, the ratio of their speeds is:
(a) 3 : 2 (b) 5 : 4
(c) 9 : 4 (d) 9 : 13

14. A train of 320 m cross a platform in 24 seconds at the speed of 120 km/h. while a man cross same platform in 4 minute. What is the speed of man in m/s?
(a) 2.4 (b) 1.5
(c) 1.6 (d) 2.0

15. A car travel first 39 km distance in 45 minute while next 25 km distance in 35 minutes. What is its average speed?
(a) 45 Km/h (b) 35 Km/h
(c) 48 Km/h (d) 90 Km/h

16. A 280 m long train moving with an average speed of 108 km/h cross a platform in 12 seconds. A boy crosses the same platform in 10 seconds. What is the speed of boy in m/s?
(a) 5 m/s (b) 8 m/s
(c) 10 m/s (d) Cannot be determined

17. A truck cover 224 km in 4 hours, the average speed of a bike is $\frac{1}{4}$ th the average speed of the truck how much distance will the bike cover in seven hour?
(a) 96 km (b) 98 km
(c) 95 km (d) 92 km

18. If a person walks at 14 km/h instead of 10 km/h he would have walked 20 km more. The actual distance travelled by him is:
(a) 85 Km (b) 50 Km
(c) 80 Km (d) 70 Km

19. Train fare between Patna to Munger for one adult is three times the train fair of one child. If adult fair is 102 then. What will be the fare of 2 adult and 3 children together for same distance?
(a) 306 (b) 212
(c) 206 (d) 214

20. A car travels a distance of 75 km at the speed of 25 km/h. It covers the next 25 km of its journey at the speed of 5 km/h and the last 50 km of its journey at the speed of 25 km/h. What is the average speed of car?
(a) 15 Km/h (b) 12.5 Km/h
(c) 40 km/h (d) 25 km/h

21. If the length of the train is 700 m and length of platform is 500 m find the time taken by the train moving at 54 km/h to cross platform?
(a) 75 sec (b) 80 sec
(c) 85 sec (d) 90 sec

22. Amit start to go to Delhi from patna at speed of 50 km/h. Distance between Delhi and Patna is 1000 km. He takes rest of 20 minutes every 3 hours of journey. How much time will he take to arrive Delhi?
(a) 20 hours (b) 21 hours
(c) 22 hours (d) 23 hours

23. A car cover a distance of 330 km in a certain amount of time at speed of 55 km/h. What is the average speed of bike that cover distance of 15 km less than car in 1 hour less than time taken by car?
(a) 50 Km/h (b) 60 Km/h
(c) 63 Km/h (d) 65 Km/h

24. A car traveling at a speed of 40 km/hour can complete a journey in 9 hours. How long will it take to travel the same distance at 60 km/hour?
(a) 6 hours (b) 3 hours
(c) 4 hours (d) $4\frac{1}{4}$ hours

25. A 75 meter long train is moving at 20 kmph. It will cross a man standing on the platform in
(a) 12 seconds (b) 14 seconds
(c) 13.5 seconds (d) 15.5 seconds

26. In what time will a train 100 meter long cross an electric pole, if its speed be 144 km/hour?
(a) 2.5 seconds (b) 5 seconds
(c) 12.5 seconds (d) 3 seconds

27. A train running at a speed of 60 kmph crosses a platform double its length in 32.4 seconds. What is the length of the platform?
(a) 180 m (b) 240 m
(c) 360 m (d) 90 m

28. A train running at the speed of 66 kmph crosses a signal pole in 18 seconds. What is the length of the train?
(a) 330 m (b) 300 m
(c) 360 m (d) 320 m

29. Two trains of equal length take 10 seconds and 15 seconds respectively to cross a telegraph post. If the length of each train be 120 metres, in what time (in seconds) will they cross each other travelling in opposite direction?
(a) 16 sec (b) 15 sec
(c) 12 sec (d) 10 sec

30. A man covers a certain distance by car driving at 70 km/hr and he returns to the starting point riding on a scooter at 55 km/hr. Find his average speed for the whole journey.
(a) 61.6 km/hr (b) 62.8 km/hr
(c) 63.6 km/hr (d) 64.6 km/hr

31. A boy walking at a speed of 10 km/hr reaches his school 15 minutes late. Next time he increases his speed by 2 km/hr, but still he is late by 5 minutes. Find the distance of his school from his house.
(a) 10 km (b) 15 km
(c) 20 km (d) 25 km

32. A motor car does a journey in 10 hrs, the first half at 21 km/hr and the second half at 24 km/hr. Find the distance?
(a) 224 km (b) 225 km
(c) 226 km (d) 228 km

33. Two men A and B starts from a place P walking at 3 km and 3.5 km an hour respectively. How many kms will they be apart at the end of 3 hrs, if they walk in opposite directions?
(a) 13.5 km (b) 15.5 km
(c) 17.5 km (d) 19.5 km

34. What is the length of the bridge which a man riding 15 km an hour can cross in 5 minutes?
(a) 850 m (b) 1050 m
(c) 1250 m (d) Can't be determined

35. A train moving with the speed of 180 km/hr. Its speed (in metres per second) is:
(a) 5 (b) 40
(c) 30 (d) 50

36. If a train, with a speed of 60 km/hr, crossed a pole in 30 second, the length of the train (in metres) is:
(a) 1000 (b) 900
(c) 750 (d) 500

37. Two trains are moving in the same direction at 50 km/hr and 30 km/hr. The faster train crossed a man in the slower train in 18 seconds. Find the length of the faster train?
(a) 80 m (b) 90 m
(c) 100 m (d) Can't be determined

38. A train running at 25 km/hr takes 18 seconds to pass a platform. Next, it takes 12 seconds to pass a man walking at 5 km/hr in the opposite direction. Find the sum of the length of the train and that of the platform?
(a) 125 m (b) 135 m
(c) 145 m (d) 155 m

39. A train is travelling at a rate of 45 km/hr. How many seconds, it will take to cover a distance of 4/5 km?
(a) 36 second (b) 64 second
(c) 90 second (d) 120 second

40. A 570 metre long train crosses a platform of equal length in 15 seconds. What is the speed of the train in metres/second?
(a) 38 (b) 54
(c) 76 (d) 70

Moderate

1. A boy is running at a speed of p kmph to cover a distance of 1 km. But, due to the slippery ground, his speed is reduced by q kmph ($p > q$). If he takes r hours to cover the distance, then :
(a) $\frac{1}{r}=\frac{1}{p}+\frac{1}{q}$ (b) $\frac{1}{r}=p-q$
(c) $r=p+q$ (d) $r=p-q$

2. A boy goes to the school with the speed of 3 kmph and returns with the speed of 2 kmph. If he takes 5 hours in all, then the distance (in kms) between the village and the school is :
(a) 6 (b) 12
(c) 8 (d) 9

3. A student walks from his house at 5 kmph and reaches his school 10 minutes late. If his speed had been 6 kmph he would have reached 15 minutes early. The distance of his school from his house is :
(a) 2.5 km (b) 12.5 km
(c) 5.5 km (d) 3.6 km

4. A train takes 18 seconds to pass completely through a station 162 m long and 15 seconds through another station 120 m long. The length of the train (in metres) is :
(a) 70 (b) 80
(c) 90 (d) 100

5. Two trains of lengths 120 m and 80 m are running in the same direction with velocities of 40 kmph and 50 kmph respectively. The time taken by them to cross each other is :
(a) 60 sec (b) 72 sec
(c) 75 sec (d) 80 sec

6. A man standing on a railway platform observes that a train going in one direction takes 4 seconds to pass him. Another train of same length going in the opposite direction takes 5 seconds to pass him. The time taken (in seconds) by the two trains to cross each other will be:
(a) $\frac{31}{9}$ (b) $\frac{40}{9}$
(c) $\frac{49}{9}$ (d) $\frac{50}{9}$

7. A train, 240m long crosses a man, walking along the line in opposite direction at the rate of 3 kmph in 10 seconds. The speed of the train is:
(a) 63 km/h (b) 75 km/h
(c) 83.4 km/h (d) 86.4 km/h

8. The distance between two cities A and B is 330km. A train starts from A at 8 am. and travels towards B at 60 km/hr. Another train starts from B at 9 am. and travels towards A at 75 km/hr. At what time do they meet?
(a) 10 am. (b) 10 : 30 am.
(c) 11 am. (d) 11 : 30 am.

9. Two men are standing on opposite ends of a bridge of 1200 metres long. If they walk towards each other at the rate of 5m/minute and 10m/minute respectively, in how much time will they meet each other?
(a) 60 minute (b) 80 minute
(c) 85 minute (d) 90 minute

10. Two trains one 160 m and the other 140 m long are running in opposite directions on parallel track, the first at 77 km an hour and the other at 67 km an hour. How long will they take to cross each other?
(a) 7 seconds (b) 7.5 seconds
(c) 6 seconds (d) 10 seconds

11. Two trains of equal length take 10 seconds and 15 seconds respectively to cross a telegraph post. If the length of each train be 120 meters, in what time (in seconds) will they cross each other travelling in same direction?

(a) 16 (b) 15
(c) 12 (d) 60

12. A person travels 600 km by train at 80km/hr, 800 km by ship at 40 km/hr. 500 km by aeroplane at 400 km/hr and 100 km by car at 50 km/hr. What is the average speed for the entire distance?

(a) $65\frac{5}{123}$ km./hr. (b) 60 km./hr.
(c) $60\frac{5}{123}$ km./hr. (d) 62 km./ hr.

13. One third of a certain journey is covered at the rate of 25 km/hr one-fourth at the rate of 30 km/hr and the rest at 50 km/hr. The average speed for the whole journey is

(a) 35 km/hr (b) $33\frac{1}{3}$ km/hr
(c) 30 km/hr (d) $37\frac{1}{12}$ km/hr

14. In covering certain distance, the speeds of A and B are in the ratio of 3 : 4 A takes 30 minutes more than B to reach the destination. The time taken by A to reach the destination is :

(a) 1 hour (b) $1\frac{1}{2}$ hours
(c) 2 hours (d) $2\frac{1}{2}$ hours

15. A car starts with the speed of 70 km/h with its speed increasing every two hours by 10 km/h. In how many hours will it cover 345 km?

(a) $2\frac{1}{4}$ hours (b) 4 hours
(c) $4\frac{1}{2}$ hours (d) 3 hour

16. Aditya rides a cycle at the speed of 15 km/h but stops for 10 minute to take rest every 20 kms. How much time will he take to cover a distance of 150 kms?

(a) 11 hours 10 minutes
(b) 11 hours
(c) 12 hours 30 minutes
(d) 15 hours

17. The ratio between the speed of bike and train is 15 : 27 respectively. Also a bus cover a distance of 720 km in 9 hours. The speed of the bike is three fourth the speed of bus. How much distance will the train cover in 7 hours?

(a) 756 km (b) 760 km
(c) 740 km (d) Cannot be determined

18. A bus started its journey from Bettiah and reached Motihari in 44 minute with its average speed 50 km/h. If the average speed of bus is increased by 5 km/h how much time will it take to cover the same distance?

(a) 31 min (b) 36 min
(c) 38 min (d) 40 min

19. A train can travel 50% faster than a bus. Both start from point A at the same time and reach point B at same time. Point B is 75 km away from point A on the way the train stopping 12.5 minutes. The speed of bus is:

(a) 110 km/h (b) 120 km/h
(c) 55 km/h (d) 60 km/h

20. A train crosses a pole in 10 seconds and a platform, which is 40% longer than the length of train in 24 seconds. If the length of platform is 140 m, what is speed of the train?

(a) 36 m/s (b) 5 m/s
(c) 10 m/s (d) 15 m/s

21. It takes eight hours for a 600 km journey if 120 km is done by train and rest by car. It takes 20 minutes more if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of car is:

(a) 2 : 3 (b) 3 : 2
(c) 4 : 3 (d) 3 : 4

22. Aditya is travelling on his cycle and has calculated that he will reach point A at 2 pm, if he travel at 10 km/h. He will reach there at 12 noon, if he travel at 15 km/h. At what speed must he travel to reach D at 1 pm?

(a) 10 kmph (b) 11 kmph
(c) 12 kmph (d) 13 kmph

23. A train running at a speed of 40 m/s crosses a pole in 21 seconds less than the time it required to cross a bridge 3.5 times its length at the same speed. What is the length of the bridge?

(a) 1080 m (b) 240 m
(c) 840 m (d) 560 m

24. Two trains A and B start at the same time from Delhi and Patna respectively towards each other they met after 16 hours. If the distance between Delhi and Patna is 1872 km and train B runs 27 km/h faster than train A what is the speed of train A?

(a) 45 km/h (b) 72 km/h
(c) 48 km/h (d) 60 km/h

25. Two trains M and N crosses a pole in 25 seconds and 1 min 15 seconds respectively. Length of train M is half length of train N. What is the respective ratio between speed of train M and train N?
(a) 2 : 3 (b) 1 : 2
(c) 2 : 1 (d) 3 : 2

26. A truck cover a distance of 396 km in 5 hour 30 minutes. The average speed of bike is $\frac{4}{3}$ of average speed of truck. What is the time taken by bike to cover distance of 12 km less than the distance which the truck cover?
(a) 4 Hrs (b) 3 Hrs
(c) 5 Hrs (d) 4 hour 15 minutes

27. Train A starts its journey from Patna to Hazipur while train B starts from Hazipur to Patna. After crossing each other they finish their journey in 81 hours and 121 hours respectively. Then what will be speed of train B if train A speed is 44 km/h?
(a) 44 Km/h (b) 55 Km/h
(c) 36 Km/h (d) 46 Km/h

28. Two train start at same time from Delhi and Mumbai and comes towards each other at the rate of 85 km and 65 km per hour respectively. When they meet it is found that one train has travelled 20 km more than the other. Find the distance between Delhi and Mumbai?
(a) 140 Km (b) 75 Km
(c) 150 Km (d) Cannot be determined

29. A train start from Delhi on Sunday at 9 am with speed of 25 km/h, another train starts from there on same day at 40 km/h at 3 pm in the same direction. Find at what distance from Delhi both train will meet?
(a) 200 Km (b) 300 Km
(c) 400 Km (d) Cannot be determined

30. An aeroplane covers a certain distance at a speed of 240 km per hour in 5 hours. To cover the same distance in $1\frac{2}{3}$ hours, it must travel at a speed of :
(a) 300 km./hr. (b) 360 km./hr
(c) 600 km./hr (d) 720 km./hr

31. A boy goes to his school from his house at a speed of 3 km/hr. If he takes 5 hours in going and coming, the distance between his house and school is:
(a) 6 km (b) 5 km
(c) 5.5 km (d) 7.5 km

32. If a train runs at 40 km/hour, it reaches its destination late by 11 minutes. But if it runs at 50 km/hour, it is late by 5 minutes only. The correct time (in minutes) for the train to complete the journey is
(a) 13 (b) 15
(c) 19 (d) 21

33. A train 800 meter long is running at the speed of 78 km/hr. if it crosses a tunnel in 1 minute, then the length of the tunnel (in meter) is :
(a) 772 (b) 500
(c) 1300 (d) 13

34. A cyclist, after cycling a distance of 70 km on the second day, finds that the ratio of distance covered by him on the first two days is 4 : 5. If he travels a distance of 42 km. on the third day, then the ratio of distances travelled on the third day and the first day is:
(a) 3 : 4 (b) 5 : 2
(c) 9 : 4 (d) 9 : 13

35. A thief is noticed by a policeman from a distance of 200m. The thief starts running and the policeman chases him. The thief and the policeman run at the rate of 10 km. and 11 km. per hour respectively. What is the distance between them after 6 minutes?
(a) 100 m (b) 190 m
(c) 200 m (d) 150 m

36. A boy started from his house by bicycle at 10 a.m. at a speed of 12 km per hour. His elder brother started after 1 hr 15 mins by scooter along the same path and caught him at 1.30 p.m. The speed of the scooter will be (in km/hr):
(a) 4.5 (b) 36
(c) $18\frac{2}{3}$ (d) 9

37. A student walks from his house at a speed of 2 km per hour and reaches his school 6 minutes late. The next day he increases his speed by 1 km per hour and reaches 6 minutes before school time. How far is the school from his house?
(a) $\frac{5}{4}$ km (b) $\frac{6}{5}$ km
(c) $\frac{9}{4}$ km (d) $\frac{11}{4}$ km

38. A moving train crosses a man standing on a platform and a bridge of 300 metres long in 10 seconds and 25 seconds respectively. What will be the time taken by the train to cross a platform 200 metres long?
(a) $16\frac{2}{3}$ seconds (b) 18 seconds
(c) 20 seconds (d) 22 seconds

39. A, B and C start at the same time in the same direction to run around a circular stadium. A completes a round in 252 seconds, B in 308 seconds and C in 198 seconds, all starting at the same point. After what time will they next meet at the starting point again?

(a) 46 minutes 12 seconds
(b) 45 minutes
(c) 42 minutes 36 seconds
(d) 26 minutes 18 seconds

40. A and B run a kilometer and A wins by 25 sec. A and C run a kilometer and A wins by 275 m. When B and C run the same distance, B wins by 30 sec. The time taken by A to run a kilometer is:
(a) 2 min 25 sec. (b) 2 min 50 sec.
(c) 3 min 20 sec. (d) 3 min 30 sec.

Difficult

1. Aditya covered a certain distance at some speed. Had he moved 3 km/h faster he would have taken 40 minute less. If he had moved 2 km/h slower he would have taken 40 minutes more. The distance is:
(a) 40 (b) 35
(c) 49 (d) 45

2. A bus take 15 hours to go from Delhi to Chandigarh. Bus go one-third time with speed of 80 km/h. 40% of rest time with speed of 70 km/h. $\frac{2}{3}$rd of rest of time with speed of 85 km/h rest with speed of 100 km/h. What is the distance that cover by bus?
(a) 1200 (b) 860
(c) 1120 (d) 1220

3. In a race of one kilometer, A gives B a start of 100 metres and still wins by 20 seconds. But if A gives B a start of 25 seconds, B wins by 50 metres. The time taken by A to run one kilometer is:
(a) $\frac{500}{29}$ seconds (b) $\frac{1200}{29}$ seconds
(c) $\frac{800}{29}$ seconds (d) $\frac{700}{29}$ seconds

4. A man covers a certain distance between his house and office on scooter. Having an average speed of 30 km/hr, he is late by 10 min. However, with a speed of 40 km/hr, he reaches his office 5 min earlier. Find the distance between his house and office.
(a) 20 km (b) 25 km
(c) 30 km (d) 35 km

5. The distance between two stations, Delhi and Amritsar, is 450 km. A train starts at 4 p.m. from Delhi and moves towards Amritsar at an average speed of 60 km/hr. Another train starts from Amritsar at 3.20 p.m. and moves towards Delhi at an average speed of 80 km/hr. How far from Delhi will the two trains meet and at what time?
(a) 5.30 p.m. (b) 5.50 a.m.
(c) 6.50 p.m. (d) 6.30 a.m.

6. A, B and C can walk at the rates of 3, 4 and 5 km per hour respectively. They starts from Pune at 1, 2, 3 o'clock respectively. When B catches A, B sends him back with a message to C. When will C get the message?
(a) 4.15 o'clock (b) 5.15 o'clock
(c) 6.25 o'clock (d) Can't be determined

7. Two trains start at the same time from two stations and proceed towards each other at the rates of 20 km and 25 km per hour respectively. When they meet, it is found that one train has travelled 80 km more than the other. Find the distance between the two stations.
(a) 720 km (b) 740 km
(c) 760 km (d) 780 km

8. A goods train and a passenger train are running on parallel tracks in the same direction. The driver of the goods train observes that the passenger train coming from behind overtakes and crossed his train completely in 60 seconds. Whereas a passenger on the passenger train marks that he crossed the goods train in 40 seconds. If the speed of the train be in the ratio of 1 : 2, find the ratio of their lengths.
(a) 1 : 2 (b) 2 : 1
(c) 3 : 2 (d) 2 : 3

9. A train after travelling 50 km meets with an accident and then proceeds at $\frac{3}{4}$ of its former speed and arrives at its destination 35 minutes late. Had the accident occurred 72 km further, it would have reached the destination only 15 minutes late. The normal speed of the train is
(a) 36 km/hr. (b) 38 km/hr.
(c) 46 km/hr. (d) 72 km/hr.

10. Two trains measuring 100 m and 80 m respectively, run on parallel lines of track. When travelling in opposite directions they are observed to pass each other in 9 seconds, but when they are running in the same direction at the same rate as before, the faster train passed the other in 18 seconds. Find the speed of the two trains in km per hour.
(a) 12 km/hr, 5 km/hr
(b) 14 km/hr, 18 km/hr
(c) 16 km/hr, 54 km/hr
(d) 18 km/hr, 54 km/hr

11. A 300 meters long train is travelling with the speed of 45 km/hr when it passes point A completely. At the same time, a motorbike starts from point A with the speed of 70 km/hr. When it exactly reaches the middle point of the train, the train increases its speed to 60 km/hr and motorbike reduces its speed to 65 km/hr. How much distance will the motorbike travel while passing the train completely?

(a) 3.8 km (b) 3.2 km
(c) 2.37 km (d) 2.2 km

12. Two trains, Kanpur Mail and Delhi Mail, start at the same time from stations Kanpur and Delhi respectively towards each other. After passing each other, they take 12 hours and 3 hours to reach Delhi and Kanpur respectively. If the Kanpur Mail is moving at the speed of 48 km/hr, the speed of the Delhi Mail is:

(a) 90 km/hr (b) 96 km/hr
(c) 86 km/hr (d) 84 km/hr

13. Heena and Renu are competing in a 100 meters race. Initially, Heena runs at twice the speed of Renu for the first fifty meters. After the 50 meters mark, Heena runs at $\frac{1}{4}$th her initial speed while Renu continues to run at her original speed. If Renu catches up with Heena at a distance of N meters from the finish line, then N is equal to:

(a) 75 meters (b) 55 meters
(c) 25 meters (d) 30 meters

14. A horse rider travels on horse back from Bhopal to Chandigarh at a constant speed. If the horse increased its speed by 6 km/hr, it would take the rider 4 hours less to cover that distance. And travelling with a speed 6 km/hr lower than the initial speed, it would take him 10 hours more than the time he would have taken if he had travelled at a speed 6 km/hr higher than the initial speed. Find the distance between Bhopal and Chandigarh?

(a) 720 km (b) 680 km
(c) 560 km (d) 480 km

15. Seeta and Geeta cycled towards each other, one from point A and the other from point B, respectively. Seeta left point A 6 hours later than Geeta left point B, and it turned out on their meeting that Seeta had travelled 12 km less than Geeta. After their meeting, they kept cycling with the same speed, and Seeta arrived at B 8 hours later and Geeta arrived at A 9 hours later. Find the speed of the faster cyclist.

(a) 3 km/hr (b) 6 km/hr
(c) 9 km/hr (d) 12 km/hr

16. Two cyclists started simultaneously towards each other and meet each other 3 hours 20 minutes later. How much time will it take the slower cyclist to cover the whole distance if the first arrived at the place of departure of the second 5 hours later than the second arrived at the point of departure of the first?

(a) 5 hours (b) 10 hours
(c) 15 hours (d) 20 hours

17. A dog finds a cat at 25 leaps away. The cat sees the dog coming towards it and starts running with the dog in hot pursuit. In every minute, the dog makes 5 leaps and the cat makes 6 leaps and one leap of the dog is equal to 2 leaps of the cat. Find the time in which the dog will catch the cat.

(a) 12.5 minutes (b) 13 minutes
(c) 11.5 minutes (d) 10.5 minutes

18. Two guns were fired from the same place at an interval of 13 minutes but a person in a train approaching the place hears the second shot 12 minutes 30 seconds after the first. Find the speed of the train, supposing that sound travels at 330 metres per second.

(a) $47\frac{13}{25}$ km / hr (b) $47\frac{11}{25}$ km / hr
(c) $47\frac{9}{25}$ km / hr (d) $47\frac{7}{25}$ km / hr

19. A man walks from A to B and back in a certain time at the rate of $3\frac{1}{2}$ km per hour. But if he had walked from A to B at the rate of 3 km an hour and back from B to A at the rate of 4 km an hour, he would have taken 5 minutes longer. Find the distance between A and B.

(a) 14 km (b) 12 km
(c) 6 km (d) 7 km

20. A carriage driving in a fog passed a man who was walking at the rate of 3 km an hour in the same direction. He could see the carriage for 4 minutes and it was visible to him upto a distance of 100m. What was the speed of the carriage?

(a) $5\frac{1}{2}$ km per hour (b) $4\frac{1}{2}$ km per hour
(c) $3\frac{1}{2}$ km per hour (d) $8\frac{1}{2}$ km per hour

21. A man leaves a point P and reaches the point Q in 4 hours. Another man leaves the point Q, 2 hours earlier and reaches the point P in 4 hours. Find the time in which the first man meets to the second man.

(a) 2 hours (b) 3.5 hours
(c) 1 hours (d) 3 hours

22. A man travels 360 km in 4 hrs, partly by air and partly by train. If he had travelled all the way by air, he would have saved $\frac{4}{5}$ of the time he was in train and would have arrived at his destination 2 hours early. Find the distance he travelled by train?

(a) 75 km (b) 90 km
(c) 85 km (d) 80 km

23. One aeroplane started 30 minutes later than the scheduled time from a place 1500 km away from its destination. To reach the destination at the scheduled time the pilot had to increase the speed by 250 km/hr. What was the speed of the aeroplane per hour during the journey?

(a) 750 km/hr (b) 755 km/hr
(c) 760 km/hr (d) 745 km/hr

24. A train leaves the station 1 hour before the scheduled time. The driver decreases its speed by 4 km/hr. At the next station 120 km away, the train reached on time. Find the original speed of the train.

(a) 24 km/hr. (b) 25 km/hr.
(c) 26 km/hr. (d) 27 km/hr.

25. A hare sees a dog 100 metres away from her and run off in the opposite direction at a speed of 12 km an hour. A minute later the dog perceives her and gives chase at a speed of 16 km per hour. How much distance cover by hare after being spotted?

(a) 750 metres (b) 800 metres
(c) 850 metres (d) 900 metres

Previous Year (Memory Based)

1. A train 180 m long moving at the speed of 20m/sec. overtakes a man moving at a speed of 10 m/sec in the same direction. The train passes the man in:

(a) 6 sec (b) 9 sec
(c) 18 sec (d) 27 sec

2. How many seconds will a 500 metre long train take to cross a man walking with a speed of 3 km/hr. in the direction of the moving train if the speed of the train is 63 km/hr?

(a) 25 sec (b) 30 sec
(c) 40 sec (d) 45 sec

3. The length of a train and that of a platform are equal. If with a speed of 90 km/hr the train crosses the platform in one minute, then the length of the train (in metres) is:

(a) 500 (b) 600
(c) 750 (d) 900

4. A train passes two bridges of length 800 m and 400 m in 100 seconds and 60 seconds respectively. The length of the train is:

(a) 80 m (b) 90 m
(c) 200 m (d) 150 m

5. Two trains are running with speeds 30 km/hr and 58 km/hr in the same direction. A man in the slower train passes by the faster train in 18 seconds. The length (in metres) of the faster train is:

(a) 70 m (b) 100 m
(c) 128 m (d) 140 m

6. Two trains are moving on two parallel tracks but in opposite directions. A person sitting in the train moving at the speed of 80 km/hr passes the second train in 18 seconds. If the length of the second train is 1000 m, its speed is:

(a) 100 km/hr (b) 120 km/hr
(c) 140 km/hr (d) 150 km/hr

7. Two trains travel in the same direction at the speeds of 56 km/h and 29 km/h respectively. The faster train passes a man in the slower train in 10 seconds. The length of the faster train (in metres) is:

(a) 100 (b) 80
(c) 75 (d) 120

8. A man standing on a platform finds that a train takes 3 seconds to pass him and another train of the same length moving in the opposite direction, takes 4 seconds. The time taken by the trains to pass each other will be:

(a) $2\frac{3}{7}$ seconds (b) $3\frac{3}{7}$ seconds
(c) $4\frac{3}{7}$ seconds (d) $5\frac{3}{7}$ seconds

9. A train passes two persons walking in the same direction at a speed of 3 km/hour and 5 km/hour respectively in 10 seconds and 11 seconds respectively. The speed of the train is:

(a) 28 km/hour (b) 27 km/hour
(c) 25 km/hour (d) 24 km/hour

10. After travelling 3 hours a train meets with an accident due to this it stops for an hour. After this the train moves at 75% speed of its original speed and reaches to destination 4 hours late. If the accident would occur at 150 km ahead in the same line then the train reaches only 3.5 hours late. Then find the distance of journey and the original speed of the train?

(a) 100 km/h, 1200 km (b) 150 km/h, 1200 km
(c) 75 km/h, 1000 km (d) 125 km/h, 900 km

11. A train starts from A at 7 a.m. towards B with speed 50 km/h. Another train from B starts at 8 a.m. with speed 60 km/h towards A. Both of them meet at 10 a.m. at C. The ratio of the distance AC to BC is:

(a) 5 : 6 (b) 5 : 4
(c) 6 : 5 (d) 4 : 5

12. Two trains start from a certain place on two parallel tracks in the same direction. The speed of the trains are 45 km/hr and 40 km/hr respectively. The distance between the two trains after 45 minutes will be:

(a) 2 km 500 m (b) 2 km 750 m
(c) 3 km 750 m (d) 3 km 250 m

13. Two trains start from stations A and B and travel towards each other at speeds of 50 km/hour and 60 km/hour respectively at same time. At the time of their meeting, the second train has travelled 120 km more than the first. The distance between A and B is:

(a) 990 km (b) 1200 km
(c) 1320 km (d) 1440 km

14. A train of 24 m length runs with a speed of 250 m/s. A man in train at the tail end of the train runs with a speed of 10 m/s. When he reaches the front end he turns back with a speed of 6 m/s and this process continues. How many rounds (up and down) he will complete if the train runs 8 kms, providing that during running he will not loose contact with the train?

(a) 8 rounds (b) 4 rounds
(c) 5 rounds (d) 7 rounds

15. A train Aligarh express leaves Delhi for Aligarh at 14:30 pm and goes at the rate of 60 km an hour and an another train, Rajdhani express leaves Delhi for Aligarh at 16 : 30 pm and goes at the rate of 80 km/h. How far from Delhi will two trains meet if they are moving in the same direction?

(a) 120 km (b) 360 km
(c) 480 km (d) 500 km

16. Two trains starting at the same time from two stations 650 km apart and going in opposite direction they meet each other after 10 hours. If one starts 4 hour 20 min late from another train then they meet after 8 hours from the starting of another train. Then find the average speed of trains?

(a) 30 km/h, 35 km/h (b) 45 km/h, 20 km/h
(c) 32 km/h, 33 km/h (d) 25 km/h, 40 km/h

17. A man can see only 50 m in fog. He was walking at a speed of 4 km/hr and saw a train was coming behind him. The train was 20 m in length and disappeared from his eyes after 30 seconds. Find the speed of the train?

(a) $18\frac{2}{5}$ km/hr (b) $6\frac{3}{5}$ km/hr

(c) $8\frac{1}{5}$ km/hr (d) $4\frac{2}{5}$ km/hr

18. Two trains start from the same point simultaneously and in the same direction. The first train travels at 40 km/h, and the speed of the second train is 25 per cent more than the speed of the first train. Thirty minutes later, a third train starts from the same point and in the same direction. It overtakes the second train 90 minutes later than it overtook the first train. What is the speed of the third train?

(a) 20 km/h (b) 60 km/h
(c) 40 km/h (d) 80 km/h

19. Two trains started at the same time, one from A to B and the other from B to A. If they arrived at B and A respectively 49 hours and 81 hours after they passes each other, the ratio of the speeds of the two trains was:

(a) 2 : 1 (b) 9 : 7
(c) 7 : 9 (d) 5 : 4

20. Two trains, A and B, start from stations X and Y respectively. After passing each other, they take 4 hours 48 minutes and 3 hours 20 minutes to reach Y and X respectively. If train A is moving at 45 km/hr., then the speed of the train B is:

(a) 60 km/hr (b) 64.8 km/hr
(c) 54 km/hr (d) 37.5 km/hr

21. A train travels some distance with a uniform speed. If the speed of the train is 20 km/hr less then it takes 16 hours more. If its speed is 10 km/hr more then it takes 5 hours less during the whole journey. Then find the distance ? And also find the speed of the train and how much time will it take to travel the whole journey?

(a) 2800 km, 70 km/h, 40 hours
(b) 3500 km, 70 km/h, 40 hours
(c) 4200 km, 140 km/h, 30 hours
(d) None of these

22. A train 550 m long is running with a speed of 55 km per hour. In what time will it pass a man who is walking at 10 km per hour in the same direction in which the train is going?

(a) 42 sec (b) 33 sec
(c) 40 sec (d) 44 sec

23. Two trains are moving in the opposite direction at 24 km and 12 km/hr. The faster train crosses a man in the slower train in 2 seconds. Find the length of the faster train:

(a) 25 m (b) 30 m
(c) 20 m (d) Data inadequate

24. In covering a certain distance, the speeds of A and B are in the ratio of 3 : 4. A takes 30 minutes more than B to reach the destination. The time taken by A to reach the destination is:

(a) 1 hour (b) 1 hour 30 min
(c) 2 hours (d) 2 hours 30 min

25. A man reduces his speed to 2/3, he takes 1 hour more in walking a certain distance. The time (in hours) to cover the distance with his normal speed is:

(a) 2 hrs (b) 1 hrs
(c) 3 hrs (d) 1.5 hrs

26. A and B start at the same time with speeds of 40 km/hr and 50 km/hr respectively. If in covering the journey A takes 15 minutes longer than B, the total distance of the journey is:

(a) 46 km (b) 48 km
(c) 50 km (d) 52 km

27. A car can cover a certain distance in $4\frac{1}{2}$ hours. If the speed is increased by 5 km/hour, it would take $\frac{1}{2}$ hour less to cover the same distance. Find the slower speed of the car:

(a) 50 km/hour (b) 40 km/hour
(c) 45 km/hour (d) 60 km/hour

28. A and B started at the same time from the same place for a certain destination. B walking at $\frac{5}{6}$ of A's speed reached the destination 1 hour 15 minutes after A. B reached the destination in:

(a) 6 hours 15 minutes (b) 7 hours 15 minutes
(c) 7 hours 30 minutes (d) 8 hours 15 minutes

29. Two rifles are fired from the same place at a difference of 11 minutes 45 seconds. But a man who is coming towards the place in a train hears the second sound after 11 minutes. Find the speed of train (assuming speed of sound = 330 m/s):

(a) 72 km/h (b) 36 km/h
(c) 81 km/h (d) 108 km/h

30. An aeroplane covers a certain distance at a speed of 240 km/hour in 5 hours. To cover the same distance in $1\frac{2}{3}$ hours, it must travel at a speed of:

(a) 30 km./hr. (b) 360 km./hr.
(c) 600 km./hr. (d) 720 km./hr.

31. A car travelling at a speed of 40 km/hour can complete a journey in 9 hours. How long will it take to travel the same distance at 60 km/hour?

(a) 6 hours (b) 3 hours
(c) 4 hours (d) $4\frac{1}{2}$ hours

32. A man travelled a certain distance by train at the rate of 25 kmph. and walked back at the rate of 4 kmph. If the whole journey took 5 hours 48 minutes, the distance was:

(a) 25 km (b) 30 km
(c) 20 km (d) 15 km

33. A boy goes to his school from his house at a speed of 3 km/hr and returns at a speed of 2 km/hr. If he takes 5 hours in going and coming, the distance between his house and school is:

(a) 6 km (b) 5 km
(c) 5.5 km (d) 6.5 km

34. A car completes a journey in 10 hours. If it covers half of the journey at 40 kmph and the remaining half at 60 kmph, the distance covered by car is:

(a) 400 km (b) 480 km
(c) 380 km (d) 300 km

35. Two cars start at the same time from one point and move along two roads at right angles to each other. Their speeds are 36 km/hour and 48 km/hour respectively. After 15 seconds the distance between them will be:

(a) 400 m (b) 150 m
(c) 300 m (d) 250 m

36. A runs twice as fast as B and B runs thrice as fast as C. The distance covered by C in 72 minutes, will be covered by A in:

(a) 18 minutes (b) 24 minutes
(c) 16 minutes (d) 12 minutes

37. The speeds of two trains are in the ratio 6 : 7. If the second train runs 364 km in 4 hours, then the speed of first train is:

(a) 60 km/hr (b) 72 km/hr
(c) 78 km/hr (d) 84 km/hr

38. A truck covers a distance of 550 metres in 1 minute whereas a bus covers a distance of 33 kms in 45 minutes. The ratio of their speeds is:

(a) 4 : 3 (b) 3 : 5
(c) 3 : 4 (d) 50 : 3

39. Three cars travelled distances in the ratio 1 : 2 : 3. If the ratio of the time of travel is 3 : 2 : 1, then the ratio of their speeds is:

(a) 3 : 9 : 1 (b) 1 : 3 : 9
(c) 1 : 2 : 4 (d) 4 : 3 : 2

40. If a man walks 20 km at 5 km/hr, he will be late by 40 minutes. If he walks at 8 km per hr, how early from the fixed time will he reach?

(a) 15 minutes (b) 25 minutes
(c) 50 minutes (d) $1\frac{1}{2}$ hours

41. In covering a distance of 30 km, Abhay takes 2 hours more than Sameer. If Abhya doubles his speed, then he would take 1 hour less than sameer. Abhay's speed (in km/hr) is:

(a) 5 (b) 6
(c) 6.25 (d) 7.5

42. A bus meets with an auto at 10 : 00 am while going on the same way in the same direction towards Haridwar. The Bus reach at Haridwar at 12 : 30 p.m. and take 1 hour rest at there. Bus return back on the same way and meet with the same auto half an hour later. At what time the Auto will reach at Haridwar:

(a) 3 pm (b) 4 pm
(c) 3 : 30 pm (d) 5 pm

43. Two swimmers started simultaneously from the beach, one to the south and the other to the East. Two hours later, the distance between them turned out to be 100 km. Find the speed of the faster swimmer, knowing that the speed of one of them was 75% of the speed of the other:
(a) 30 kmph (b) 40 kmph
(c) 45 kmph (d) 60 kmph

44. A constable is 114 metres behind a thief. The constable runs 21 metres and the thief 15 metres in a minute, IN what time will the constable catch the thief?
(a) 19 minutes (b) 18 minutes
(c) 17 minutes (d) 16 minutes

45. A thief, who was chased by a policeman was 100 metres ahead of the policeman initially. If the ratio between speeds of the policeman and the thief is 5 : 4, then how long thief would have covered before he was caught by the policeman?
(a) 80 m (b) 200 m
(c) 400 m (d) 600 m

46. Ram covered a certain distance at some speed. Had he moved 3 km per hour faster, he would have taken 40 minutes less. If he had moved 2 km per hour slower, he would have taken 40 minutes more. The distance (in km) is:
(a) 20 km (b) 35 km
(c) $36\frac{2}{3}$ km (d) 40 km

47. When a bus stops at different stations then its average speed is 40 km/h. But when it travels without stoppage the average speed of the bus is 50 km/h. Then find how many minutes the bus stops in an hour?
(a) 12 min (b) 15 min
(c) 25 min (d) 10 min

48. A boy rides his bicycle 10 km at an average speed of 12 km per hr and again travels 12 km at an average speed of 10 km per hr. His average speed for the entire trip is approximately:
(a) 10.4 km/hr (b) 10.8 km/hr
(c) 11.0 km/hr (d) 12.2 km/hr

49. A train moves with a speed of 30 kmph for 12 minutes and for next 8 minutes at a speed of 45 kmph. Find the average speed of the train:
(a) 37.5 kmph (b) 36 kmph
(c) 48 kmph (d) 30 kmph

50. A man completed a certain journey by a car. If he covered 30% of the distance at the speed of 20 km/hr, 60% of the distance at 40 km/hr and the remaining distance at 10 km/hr his average speed for the whole journey was:
(a) 25 km/hr (b) 28 km/hr
(c) 30 km/hr (d) 33 km/hr

Foundation

Solutions

1. **(c);** Average speed $= \dfrac{50+40+90}{\frac{50}{25}+\frac{40}{20}+\frac{90}{15}}$

$= \dfrac{180}{2+2+6} = \dfrac{180}{10} = 18$ kmph

2. **(c);** Average speed= $\dfrac{\text{total distance}}{\text{total time}} = \dfrac{5\times6+4\times12}{6+12}$

$= \dfrac{30+48}{18} = \dfrac{78}{18} = \dfrac{13}{3} = 4\dfrac{1}{3}$ km/hr

3. **(b);** Speed $= \dfrac{\text{distance}}{\text{time}} = \dfrac{10\frac{1}{5}}{3} = \dfrac{51}{5}\times\dfrac{1}{3} = \dfrac{17}{5}$ km/hr

$\therefore$ Required distance = speed × time

$= \dfrac{17}{5}\times5 = 17$ km

4. **(d);** Speed of the train $= \dfrac{110}{3}$ m/sec.

Required time $= \dfrac{(165+110)}{110}\times3$

$= \dfrac{275}{110}\times3 = 7.5$ sec.

5. **(d);** (Length of the 1st train + Length of the tunnel) = speed × time

$x + 700 = \dfrac{72\times5}{18}\times(1\times60)$

$x + 700 = 20 \times 60$

$x + 700 = 1200 \Rightarrow x = 1200 - 700$

$x = 500$ metres.

6. **(c);** Speed of the train $= \dfrac{200+200}{20}$

$= \dfrac{400}{20} = 20$ m/sec $= 20\times\dfrac{18}{5}$ km/hr

$= 72$ km/hr

7. (b); Since the trains are moving in opposite direction

$\therefore$ Relative speed = speed of first train + speed of second train

Let speed of second train = x km/hr

$\therefore (65+x)\times\frac{5}{18}=\frac{125+125}{6}, (65+x)\times\frac{5}{18}=\frac{250}{6}$

(65 + x) = 150 $\Rightarrow$ x = 150 – 65 $\Rightarrow$ x = 85 km/hr.

8. (c); Let usual time = t, distance = d, and speed = s

$\therefore s=\frac{d}{t}$ ... (i)

and $\frac{3}{5}s=\frac{d}{t+2\frac{1}{2}}$ ... (ii)

From equation (i) ÷ equation (ii)

$\frac{s}{\frac{3}{5}s}=\frac{\frac{d}{t}}{\frac{d}{t+\frac{5}{2}}} \Rightarrow \frac{5}{3}=\frac{t+\frac{5}{2}}{t} \Rightarrow 5t=3t+\frac{15}{2}$

$2t=\frac{15}{2} \Rightarrow t=\frac{15}{4}=3\frac{3}{4}$ hours

9. (c); At normal speed, Let usual time = t

$\therefore s=\frac{d}{t}$ ... (i)

and $\frac{7}{11}s=\frac{d}{22}$ ... (ii)

From equation (i) ÷ equation (ii)

$\frac{s}{\frac{7}{11}s}=\frac{\frac{d}{t}}{\frac{d}{22}} \Rightarrow \frac{11}{7}=\frac{22}{t} \Rightarrow t=14$ hours.

$\therefore$ saved time = (22 – 14) hrs = 8 hrs.

10. (c); Let required time = t hours;

$\therefore s=\frac{d}{t}$ ... (i)

$\frac{3}{4}s=\frac{d}{(t+2)}$ ... (ii)

$\frac{s}{\frac{3}{4}s}=\frac{\frac{d}{t}}{\frac{d}{(t+2)}} \Rightarrow \frac{4}{3}=\frac{t+2}{t}$

4t = 3t + 6 $\Rightarrow$ t = 6 hours

11. (a); Average speed $=\frac{2xy}{x+y}=\frac{2\times12\times18}{(12+18)}$

$=\frac{72}{5}=14\frac{2}{5}$ km/hr

12. (b); From the formula –

$\therefore$ Required ratio = $\sqrt{b}:\sqrt{a}=\sqrt{9}:\sqrt{4}=3:2$

13. (b); Required ratio = $\sqrt{b}:\sqrt{a}$

$=\sqrt{25}:\sqrt{16}=5:4$

14. (d); Let length of the platform = x m

$\therefore 120\times\frac{5}{18}=\frac{320+x}{24}$

800 = 320 + x $\Rightarrow$ x = 480 m

$\therefore$ Speed of the man $=\frac{480}{4\times60}=\frac{120}{60}=2.0$ m/sec.

15. (c); Average speed

$=\frac{39+25}{(45+35)\times\frac{1}{60}}=\frac{64\times60}{80}=48$ km/hr

16. (b); Let length of the platform = x

$\therefore 108\times\frac{5}{18}=\frac{(280+x)}{12} \Rightarrow 360=280+x$

x = 80 metres

$\therefore$ Speed of Boy $=\frac{80}{10}=8$ m/sec

17. (b); Speed of the truck $=\frac{224}{4}=56$ km/hr

$\therefore$ average speed of the bike $=\frac{56}{4}=14$ km/hr

$\therefore$ Required distance = 14 × 7 = 98 km

18. (b); When person walks at 14 km/h

$\therefore 14=\frac{d}{t}$ ... (i)

Now speed = 10 km/h

$10=\frac{(d+20)}{t}$ ... (ii)

$\therefore$ From eqn(i) ÷ eqn(ii)

$\frac{14}{10}=\frac{\frac{d}{t}}{\frac{(d+20)}{t}}$

$\frac{14}{10}=\frac{d}{d+20} \Rightarrow 14d+280=10d \Rightarrow 4d=280$

d = 70km

$\therefore$ Actual distance = 70 – 20 = 50 km

19. (a); Given adult fair = 102

$\therefore$ Child fair $= \frac{102}{3} = 34$

Required fair $= 2(102) + 3(34) = 204 + 102 = 306$

20. (a); Average speed

$= \frac{\text{total distance}}{\text{total time}} = \frac{75+25+50}{\frac{75}{25}+\frac{25}{5}+\frac{50}{25}}$

$= \frac{150}{3+5+2} = \frac{150}{10} = 15\text{ km/hr}$

21. (b); Speed $= \frac{\text{distance}}{\text{time}}$, $54 \times \frac{5}{18} = \frac{700+500}{t}$

$t = \frac{1200}{15}\text{ sec} \Rightarrow t = 80\text{ sec.}$

22. (c); Running time $= \frac{1000}{50} = 20$ hrs.

No. of times that he will rest $= \frac{20}{3} \approx 6$

$\therefore$ Rest time = 6 × 20 = 120 min = 2 hours

$\therefore$ Total time = (20 + 2) = 22 hrs.

23. (c); Time taken by car $= \frac{330}{55} = 6$ hrs.

$\therefore$ Average speed of bike $= \frac{(330-15)}{(6-1)} = \frac{315}{5}$

= 63 km/hr

24. (a); distance = 40 × 9 = 360 km

Required time $= \frac{360}{60} = 6$ hrs.

25. (c); Speed $= \frac{\text{distance}}{\text{time}} \Rightarrow 20 \times \frac{5}{18} = \frac{75}{t}$

$t = \frac{75 \times 18}{100} = \frac{54}{4} = 13.5$ seconds

26. (a); Let Required time = t

$144 \times \frac{5}{18} = \frac{100}{t} \Rightarrow t = 2.5$ seconds

27. (c); Speed $= \frac{\text{distance}}{\text{time}}$

Let length of the platform = 2x m

$\frac{60 \times 5}{18} = \frac{x+2x}{32.4} \Rightarrow 3x = 1.8 \times 60 \times 5$

3x = 90 × 6

x = 30 × 6 = 180 m

$\therefore$ Length of the plat form = 2x = 360 m

28. (a); Speed $= \frac{\text{distance}}{\text{time}} \Rightarrow 66 \times \frac{5}{18} = \frac{d}{18}$

d = 330 m

29. (c); Speed of first train $(S_1) = \frac{120}{10} = 12$ m/sec.

Speed of second train $(S_2) = \frac{120}{15} = 8$ m/sec.

$\therefore$ Required time $= \frac{120+120}{(12+8)} = \frac{240}{20} = 12$ sec

30. (a); Average speed $= \frac{2xy}{x+y} = \frac{2 \times 70 \times 55}{125}$

= 61.6 km/hr

31. (a); Let distance = d, time = t

$\therefore 10 = \frac{d}{t + \frac{15}{60}}$ ...(i)

$12 = \frac{d}{t + \frac{5}{60}}$ ...(ii)

From equation (i) ÷ equation (ii)

$\frac{10}{12} = \frac{t + \frac{1}{12}}{t + \frac{1}{4}} \Rightarrow \frac{10}{12} = \frac{4 \times (12t+1)}{12(4t+1)} \Rightarrow t = \frac{3}{4}\text{hr}$

$d = 10 \times \left(\frac{3}{4} + \frac{15}{60}\right) = 10 \times \frac{60}{60} = 10\text{km}$

Shortcut:

Let distance = d

$\frac{d}{10} - \frac{d}{12} = \frac{10}{60} \Rightarrow d = 10\text{ km}$

32. (a); Average speed $= \frac{2xy}{x+y} = \frac{2 \times 21 \times 24}{45}$

$= \frac{112}{5}$ km / hr

$\therefore$ Distance $= \frac{112}{5} \times 10$ km = 224 km

33. (d); Relative speed = (3 + 3.5) km/hr = 6.5 km/hr

time = 3 hrs

$\therefore$ required distance = (6.5×3) = 19.5 km

34. (c); Let length of the bridge = x

$\therefore$ speed $=\frac{\text{distance}}{\text{time}} \Rightarrow 15=\frac{x}{5}\times 60$

$x=\frac{75}{60}\text{km} =\frac{5}{4}\text{km} =\frac{5}{4}\times 1000 \Rightarrow x = 1250$ m

35. (d); Speed in metres per second

$=180\times\frac{5}{18}=50$ m/sec.

36. (d); Speed $=\frac{\text{distance}}{\text{time}} \Rightarrow 60\times\frac{5}{18}=\frac{d}{30} \Rightarrow d = 500$ m

37. (c); Relative speed of the trains

$= (50 - 30) = 20$ km/hr $= 20\times\frac{5}{18}$ m / sec.

$\therefore$ length of the faster train $=20\times\frac{5}{18}\times 18=100$ m

38. (a); Let length of the train = l_1, length of platform = l_2

$\therefore (25+5)\times\frac{5}{18}=\frac{l_1}{12} \Rightarrow \frac{150}{18}=\frac{l_1}{12} \Rightarrow l_1 = 100$m

$\therefore 25\times\frac{5}{18}=\frac{100+l_2}{18} \Rightarrow 100+l_2=125 \Rightarrow l_2 = 25$m

$\therefore$ Required length (100 + 25) = 125 m

39. (b); Time $=\frac{4}{5}\times\frac{1}{45}$ hrs

$=\frac{4}{5}\times\frac{1}{45}\times 60\times 60$ sec = 64 seconds

40. (c); Speed of the train $=\frac{570+570}{15}$

$=\frac{1140}{15}=76$ m/sec.

Moderate

1. (b); By the given statements

Speed $=\frac{\text{Distance}}{\text{Time}} \Rightarrow p-q=\frac{1}{r} \Rightarrow \frac{1}{r}=p-q$

2. (a); Let distance = d

$\frac{d}{2}+\frac{d}{3}=5 \Rightarrow \frac{5d}{6}=5 \Rightarrow d = 6$ km

3. (b); Let distance = d

$\frac{d}{5}-\frac{d}{6}=\frac{25}{60}$

$\frac{d}{30}=\frac{25}{60} \Rightarrow d = 12.5$ km

4. (c); Let length of train = L

$\therefore$ speed $=\frac{L+162}{18}$ ... (i)

speed $=\frac{L+120}{15}$ ...(ii)

From equation (i) and equation (ii)

$\frac{L+162}{18}=\frac{L+120}{15}$

5L + 810 = 6L + 720 $\Rightarrow$ L = 90

5. (b); Relative speed of two train = (50 – 40)

= 10 kmph $=10\times\frac{5}{18}$ m/s

Required time $=\frac{120+80}{50}\times 18=\frac{200}{50}\times 18 = 72$ sec.

6. (b); Let length of each train = x metres

$\therefore$ speed of first train $=\frac{x}{4}$ m/sec

speed of second train $=\frac{x}{5}$ m/sec

Relative speed $=\frac{x}{4}+\frac{x}{5}=\frac{9x}{20}$ m/sec

Time taken to cross each other
= time train to cover 2x metres

at $\left(\frac{9x}{12}\right)$ m/sec $=2x\times\frac{20}{9x}=\frac{40}{9}$ sec.

7. (c); Let the speed of the train = s km/hr

$(s+3)\times\frac{5}{18}=\frac{240}{10} \Rightarrow 24\times 18 = 5s + 15$

432 = 5s + 15 $\Rightarrow$ 5s = 417 $\Rightarrow$ s = 83.4 km/hr

8. (c); Let the time taken by train A when it meet = t
Let the time taken by train B when it meet = (t – 1)
60 × t + 75 (t – 1) = 330 $\Rightarrow$ 60t + 75t – 75 = 330
135 t = 405 $\Rightarrow$ t = 3 hr
$\therefore$ both train will meet at 11 A.M.

9. (b); Let they meet in t min.
5t + 10t = 1200
15t = 1200
t = 80 min

10. (b); Relative speed = (77 + 67) km/hr = 144 km/hr

$=\frac{144\times 5}{18}$ m/sec = 8 × 5 = 40 m/sec.

$\therefore$ Required time $=\frac{(160+140)}{40}=\frac{300}{40}=7.5$ sec.

11. (d); Speed of first train $(S_1) = \frac{120}{10} = 12\ m/sec$

Speed of second train $(S_2) = \frac{120}{15} = 8\ m/sec$

$\therefore$ Relative speed $= S_1 - S_2 = 4\ m/sec$

$\therefore$ Required time $= \frac{120+120}{4} = \frac{240}{4} = 60$ sec

12. (a); Average speed $= \frac{\text{total distance}}{\text{total time}}$

$$= \frac{2000}{\frac{3000+8000+500+800}{400}}$$

$$= \frac{2000 \times 400}{12300} = \frac{8000}{123} = 65\frac{5}{123}\ km/hr$$

13. (b); Let total distance = d km

$\therefore$ average speed $= \frac{\text{total distance}}{\text{total time}}$

$$= \frac{d}{\frac{d}{3}\times\frac{1}{25}+\frac{d}{4}\times\frac{1}{30}+\frac{5d}{12}\times\frac{1}{50}}$$

$$= \frac{1}{\frac{1}{75}+\frac{1}{120}+\frac{5}{600}} = \frac{1}{\frac{8+5+5}{600}}$$

$$= \frac{600}{18} = \frac{300}{9} = \frac{100}{3} = 33\frac{1}{3}\ km/hr$$

14. (c); Let speed of A = 3x

Let speed of B = 4x

$3x \times t = 4x\left(t - \frac{30}{60}\right) \Rightarrow 3t = 4t - 2 \Rightarrow t = 2$ hrs.

15. (c); In first two hours the distance covered by car = 70 + 70 = 140 km

In the four hours distance covered by car = 140 + 160 = 300

Remaining 45km is covered by car in

$= \frac{45}{90}$ hr $= \frac{1}{2}$ hr; $\therefore$ Total time $= 4\frac{1}{2}$ hrs.

16. (a); No. of stops that he rests $= \frac{150}{20} \approx 7$

$\therefore$ rest time = 7 × 10 = 70 min = 1 hr 10 min

$\therefore$ run time $= \frac{150}{15} = 10$ hrs

$\therefore$ Total time = 11 hr 10 min

17. (a); Let speed of bike = 15x

Speed of train = 27x

$\therefore$ speed of bus $= \frac{720}{9} = 80\ km/hr$

$15x = \frac{3}{4}(80) \Rightarrow 15x = 60 \Rightarrow x = 4$

Speed of train = 108 km/hr

$\therefore$ Required distance = (108 × 7) km = 756 km

18. (d); Distance $= \frac{44}{60} \times 50$ km

$\therefore$ time when speed of bus increased by

$5\ km/hr = \frac{44}{60} \times 50 \times \frac{1}{55} = \frac{4}{6}$ hrs $= \frac{4}{6} \times 60$ min.

= 40 min

19. (b); Let speed of bus = 2x

$\therefore$ speed of train = 3x

$\frac{75}{2x} - \frac{75}{3x} = \frac{12.5}{60} \Rightarrow \frac{125}{600} = \frac{75}{x}\left(\frac{1}{2} - \frac{1}{3}\right)$

$= \frac{75}{x}\left(\frac{1}{6}\right) \Rightarrow \frac{125}{100} = \frac{75}{x}$

$x = \frac{75 \times 100}{125} \Rightarrow x = 4 \times 15 \Rightarrow x = 60\ km/hr$

$\therefore$ speed of bus = 2 × 60 = 120 km/hr

20. (c); Length of platform = 140 m

Let length of train = x m

$x + x \times \frac{40}{100} = 140 \Rightarrow x = 100$ m

Speed $= \frac{100}{10} = 10\ m/sec.$

21. (d); Let speed of train = x

Let speed of car = y

$\frac{120}{x} + \frac{480}{y} = 8$ hours ... (i)

$\frac{200}{x} + \frac{400}{y} = 8\frac{1}{3}$ ... (ii)

equation (i) × 5 – equation (ii) × 3

$\frac{600}{x} + \frac{2400}{y} = 40 \Rightarrow \frac{600}{x} + \frac{1200}{y} = 25$

On subtracting (2) from (1)

$\frac{1200}{y} = 15 \Rightarrow y = 80 \Rightarrow \frac{120}{x} + 6 = 8$

$\frac{120}{x} = 2 \Rightarrow x = 60$

$\therefore$ Require ratio = x : y = 60 : 80 = 6 : 8 = 3 : 4

22. (c); Let Aditya takes t hours when he travel at 10 km/hr

$$\frac{d}{10}-\frac{d}{15}=2 \Rightarrow \frac{d}{30}=2 \Rightarrow d=60\text{ km}$$

$$\therefore\ t=\frac{60}{10}=6\text{ hr}$$

$\therefore$ to reach A at 1 pm, speed $=\frac{60}{5}=12$ kmph

23. (c); Let length of train = x m

$\therefore$ Let length of bridge = 3.5x m

$$\frac{x+3.5x}{40}-\frac{x}{40}=21 \Rightarrow x+3.5x-x=40\times 21$$

$$3.5x=840 \Rightarrow x=\frac{840}{35}\times 10$$

$\therefore$ Length of bridge $=\frac{840}{7}\times 2\times\frac{7}{2}=840$ m

24. (a); Let speed of train A = x km/hr

then Let speed of train B = (x + 27) km/hr

$\therefore$ 16x + 16(x + 27) = 1872

16x + 16x = 1872 – 432

32x = 1440 $\Rightarrow$ x = 45 km/hr

25. (d); Let length of train N = x

$\therefore$ Let length of train M $=\frac{x}{2}$

$\therefore$ speed of train M $=\frac{x}{2\times 25}=\frac{x}{50}$ m/s

Speed of train N $=\frac{x}{75}$ m/s

Required ratio $=\frac{x}{50}:\frac{x}{75}=\frac{1}{50}:\frac{1}{75}=3:2$

26. (a); Speed of truck $=\frac{396}{11}\times 2=36\times 2=72$ km/hr

Speed of bike $=\frac{4}{3}\times 72=24\times 4=96$ km/hr

Required time $=\frac{396-12}{96}=\frac{384}{96}=\frac{32}{8}=4$ hrs.

27. (c); Ratio of their speed = $\sqrt{b}:\sqrt{a}=\sqrt{121}:\sqrt{81}$

= 11 : 9

Given A's speed = 44 km

$\therefore$ B's speed = 36 km/hr

28. (c); Let distance travelled by second train = d

$$\frac{d}{65}=\frac{d+20}{85}$$

85d = 65d + 65×20

20d = 65×20, d = 65 km

$\therefore$ Total distance = 65 + 65 + 20 = 150 km

29. (c); Let time taken by first train = t hrs

25t = 40(t – 6), 25t = 40t – 240

15t = 240, t = 16 hrs

$\therefore$ Required distance = 16 × 25 = 400

30. (d); Distance = 240 × 5 = 1200 km

$\therefore$ required speed $=\frac{1200}{5}\times 3=240\times 3$

= 720 km/hr

31. (d); Distance $=3\times\frac{5}{2}$ km = 7.5 km

32. (c); $40=\dfrac{d}{t+\frac{11}{60}}$ ... (i)

$50=\dfrac{d}{t+\frac{5}{60}}$ ... (ii)

By (i) and (ii)

$$\frac{4}{5}=\frac{t+\frac{5}{60}}{t+\frac{11}{60}} \Rightarrow \frac{4}{5}=\frac{60(60t+5)}{60(60t+11)}$$

240t + 44 = 300t + 25 $\Rightarrow$ 60t = 19

$$t=\frac{19}{60}\text{hr} \Rightarrow t=\frac{19}{60}\times 60\text{ min} \Rightarrow t=19\text{ min}$$

33. (b); Let length of tunnel = L

$$\therefore\ 78\times\frac{5}{18}=\frac{800+L}{60} \Rightarrow 1300=800+L \Rightarrow L=500$$

34. (a); Distance travelled by A in first day $=4\times\frac{70}{5}$

= 56 km

$\therefore$ Required ratio = 42 : 56 = 6 : 8 = 3 : 4

35. (a); Relative speed = (11 – 10) km/hr = 1 km/hr

in 6 min., distance betwen both $=\frac{1\times 5}{18}\times 6\times 60$

= 100 m

$\therefore$ Required distance = (200 – 100) m = 100 m

36. (c); At 1 : 30 PM., distance covered by boy

$=\frac{7}{2}\times 12=42$ km

Let speed of scooter = x km/hr

$$42=x\times\left(\frac{7}{2}-\frac{5}{4}\right) \Rightarrow 42=x\times\frac{14-5}{4}$$

$$42=x\times\frac{9}{4} \Rightarrow x=\frac{42\times 4}{9} \Rightarrow x=\frac{56}{3}$$

$$x=18\frac{2}{3}\text{ km/hr}$$

37. **(b);** Let distance = x km

$$\therefore \frac{x}{2}-\frac{x}{3}=\frac{12}{60} \Rightarrow \frac{3x-2x}{6}=\frac{1}{5} \Rightarrow x=\frac{6}{5}\text{km}$$

38. (c); Let length of train = L

$$\therefore s=\frac{L}{10} \quad \text{... (i)}$$

$$s=\frac{L+300}{25} \quad \text{... (ii)}$$

by equation (i) and equation (ii)

$$\frac{L}{10}=\frac{L+300}{25} \Rightarrow 25L = 10L + 3000$$

$15L = 3000 \Rightarrow L = 200$ m $\Rightarrow s = 20$ m/sec

$$\therefore \text{Required time} = \frac{200+200}{20}=\frac{400}{20}=20 \text{ seconds}$$

39. **(a);** Time taken by A = 252 = 2 × 2 × 3 × 3 ×7

Time taken by B = 308 = 2 × 2 ×7 × 11

Time taken by C = 198 = 2 × 3 × 3 × 11

∴ L.C.M = 2772 sec

= 46 min 12 sec

40. **(c);** A win by B in 25 sec

B win by C in 30 sec

A win by C in 55 sec

A win by C by 275 m

$$\text{Speed of A} = \frac{275}{55} = 5 \text{ m/s}$$

$$\text{Time taken to run a kilometre by A} = \frac{1000}{5} = 200$$

= 3 min 20 sec

Difficult

1. **(a);** Let distance = d, usual speed = s km/h

$$\frac{d}{s}-\frac{d}{s+3}=\frac{40}{60}$$

$2s(s + 3) = 9d$ ....(i)

$$\frac{d}{s-2}-\frac{d}{s}=\frac{40}{60}$$

$s(s - 2) = 3d$ ...(ii)

Solving equation (i) and equation (ii)

d = 40

2. **(d);** $\text{Average speed} = \dfrac{\text{total distance}}{\text{total time}}$

distance = speed × time

$$=\frac{15}{3}\times 80+4\times 70+4\times 85+2\times 100$$

= 400 + 280 + 340 + 200 = 1220

3. **(a);** Let A takes x seconds in covering 1000m and B takes y seconds covering the same distance

$$\therefore x+20=\frac{900}{1000}y \Rightarrow \frac{10x}{9}+\frac{200}{9}=y \quad \text{... (i)}$$

$$\frac{950}{1000}x+25=y \quad \text{...(ii)}$$

From equation (i) and (ii)

$$\Rightarrow \frac{10x}{9}+\frac{200}{9}=\frac{950x}{1000}+25 \text{ (From Eq. (i))}$$

$$\frac{200x-171x}{180}=\frac{225-200}{9}$$

$$x=\frac{25}{9}\times\frac{180}{29}=\frac{500}{29} \text{ seconds}$$

4. **(c);** Let distance = x km

$$\therefore \frac{x}{30}-\frac{x}{40}=\frac{1}{4} \Rightarrow \frac{4x-3x}{120}=\frac{1}{4} \Rightarrow x=30\text{ km}$$

5. **(c);** Let required time = t

$$60\left(t-\frac{2}{3}\right)+80t=450 \Rightarrow$$

$60t - 40 + 80t = 450 \Rightarrow 140t = 490$

$$t=\frac{7}{2} \Rightarrow t=3\frac{1}{2}$$

∴ Required time = 3 : 20 + 3 : 30 = 6 : 50 PM

6. **(b);** Let the time to catch A for B = t

∴ $3(t + 1) = 4t \Rightarrow t = 3$hrs.

∴ Distance covered by A = 3 × 4 = 12 km

distance covered by C in 2hr = 10 km

$$\therefore 2 = 3t + 5t \Rightarrow t = \frac{1}{4}\text{ hr} = 15 \text{ min}$$

∴ Required = 5 : 15 o'clock

7. **(a);** $\dfrac{d}{20}=\dfrac{d+80}{25}$

$25d = 20d + 1600 \Rightarrow d = 320$

∴ Total distance = 320 + 320 + 80 = 640 + 80

= 720 km

8. **(b);** Let length of good train = L_1, speed = S_1

length of passenger train = L_2, speed = S_2

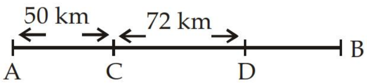
$$(S_2 - S_1)=\frac{L_1+L_2}{60} \quad \text{... (i)}$$

$$(S_2 - S_1)=\frac{L_1}{40} \quad \text{... (ii)}$$

On solving (i) and (ii)

$$\frac{L_1}{L_1+L_2}=\frac{40}{60}=\frac{2}{3},\quad \frac{L_1}{L_2}=\frac{2}{1}=2:1$$

9. (d);



(i) When accident takes place at C then the distance covered after the accident = CB

(ii) When accidnent taken place at D then distance covered after the accident = DB

CB – DB = 72 km

CD = 72 km

Let normal speed = 4x

Speed after accident = 3x

$$\frac{72}{3x}-\frac{72}{4x}=\frac{35}{60}-\frac{15}{60},\ x=18$$

So, normal speed = 72 km/h

10. (d); Let speed of two trains are S_1 and S_2

$$\therefore S_1+S_2=\frac{100+80}{9}=\frac{180}{9}=20 \quad \text{... (i)}$$

$$S_1-S_2=\frac{100+80}{18}=\frac{180}{18}=10 \quad \text{... (ii)}$$

On solving (i) and (ii)

$\therefore 2S_1 = 30$

$$S_1=15\text{ m/sec}=15\times\frac{18}{5}=54\text{km/hr}$$

$$\therefore S_2=5\text{m/sec}=5\times\frac{18}{5}=18\text{ km/hr}$$

11. (c); When the motorbike reached to the middle point Distance = 150 m, relative speed = (70 – 45) = 25 km/hr.

$$\therefore \text{Time}=\frac{150}{25\times5}\times18\text{ sec.}$$

$\therefore$ Distance covered by motorbike

$$=\frac{150}{25\times5}\times18\times70\times\frac{5}{18}=420\text{ m}$$

When it reaches exactly middle point of the train after that distance to be covered = 150 m relative speed = (65 – 60) = 5 km/hr

$$\text{Time}=\frac{150}{5\times5}\times18$$

$\therefore$ Distance covered by bike

$$=\frac{150\times8}{5\times5}\times\frac{65\times5}{18}=1950\text{ m}$$

$\therefore$ total distance = 1950 + 420 = 2370 m = 2.37 km

12. (b); $\dfrac{\text{Speed of Kanpur mail}}{\text{Speed of Delhi mail}}=\dfrac{\sqrt{3}}{\sqrt{12}}$

Speed of Delhi mail =

$$\frac{\sqrt{12}\times48}{\sqrt{3}}=2\times48=96\text{ km/h}$$

13. (c); Let initially speed of Heena = 4x

Let initially speed of Renu = 2x

$\therefore$ Time will be same when Renu catches Heena.

$$\frac{50}{4x}+\frac{d}{x}=\frac{50+d}{2x}$$

$$\frac{50+4d}{4x}=\frac{50+d}{2x}\Rightarrow 50+4d=100+2d$$

$2d = 50 \Rightarrow d = 25$

$\therefore$ Required distance (N) = 100 – (50 + 25) = 25 m

14. (a); From the first case, $\dfrac{d}{s}-\dfrac{d}{s+6}=4$ ... (i)

From the second case, $\dfrac{d}{s-6}-\dfrac{d}{s+6}=10$ ... (ii)

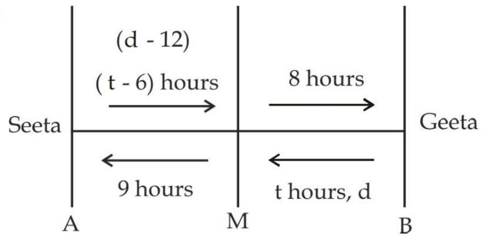
If we solve equation(i) and (ii) then it is very long and

tedious method. So here $\dfrac{d}{s},\dfrac{d}{s-6},\dfrac{d}{s+6}$ should be represent integers to satisfy both equation

$\therefore$ D should be the value which have three divisors which are 6 units apart from each other.

So only 720 satisfy the condition which have 3, 9, 15 divisors which are 6 units apart.

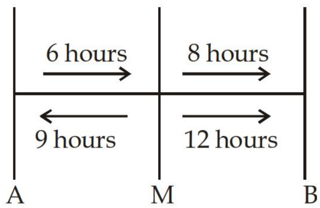
15. (b);



From the figure it is clear that seeta is faster as she takes only (t + 2) hours while Geeta has taken (t + 9) hours to complete the journey.

$$\text{Then, we get }=\frac{t-6}{9}=\frac{8}{t}\Rightarrow t^2-6t=72$$

$t^2 - 6t - 72 = 0 \Rightarrow t = 12, -6;\quad \therefore t = 12$



$\therefore$ ratio of the speed = 3 : 2

$\therefore$ 12 × 2s – 6 × 3s = 12; s = 2 km/hr

$\therefore$ Speed of faster person = 6 km/hr

16. (b); Since the two cyclist meet after = 200 minutes they cover 0.5% of the distance per minutes and 30% per hour.

This condition is satisfied only if the slower cyclist takes 10 hour (there by covering 10% per hour) and faster cycle takes 5 hours (there by covering 20% per hour.

17. (a); Initial distance = 25 dogs leaps

Per minute dog makes 5 dog leaps

Per minute cat makes 6 cat leaps = 3 dog leaps

Relative speed = 2 dogs leaps per minutes

Thus, an initial distance of 25 dogs leaps would be covered in 12.5 minutes.

18. (a); Distance travelled by train in

$12\frac{1}{2}$ min = Distance travelled by sound in 30 sec

= 330 × 30 m/sec

$\therefore$ Speed of the train $= \frac{330 \times 30}{25 \times 60} \times 2$ m/sec

$= \frac{330 \times 30}{25 \times 60} \times 2 \times \frac{18}{5}$ km/hr $= \frac{1188}{25} = 47\frac{13}{25}$ km

19. (d); Let distance = x km

In the second case average speed $= \frac{2 \times 3 \times 4}{3+4} = \frac{24}{7}$

$\therefore \frac{2x \times 7}{24} - \frac{2x \times 2}{7} = \frac{5}{60}$

$\frac{98x - 96x}{24 \times 7} = \frac{5}{60} \Rightarrow \frac{2x}{24 \times 7} = \frac{5}{60} \Rightarrow x = 7$ km

20. (b); Distance travelled by man in 4 min.

$= \frac{4}{60} \times 3 = \frac{1}{5}$ km = 200 m

Distance travelled by carriage in 4 min. = (200 + 100) = 300 min.

$\therefore$ Speed of carriage $= \frac{300}{4 \times 60}$ m/sec $= \frac{5}{4}$ m/sec

$= \frac{5}{4} \times \frac{18}{5}$ km/hr $= \frac{9}{2} = 4\frac{1}{2}$ km per hour

21. (c); Let distance = d

Let A meet B after x hours from point P

Speed of the first man $= \frac{d}{4}$

Speed of the second man $= \frac{d}{4}$

$\therefore \frac{d}{4} \times x + \frac{d}{4}(x+2) = d$

$\frac{x}{4} + \frac{x+2}{4} = 1 \Rightarrow x + x + 2 = 4$

$2x = 2 \Rightarrow x = 1$hr

22. (b); Given,

total time = 4 hrs.

Let time travelled by aeroplane = x

$\therefore$ time travelled by train = (4 – x)

$\therefore 2 = \frac{4}{5}(4 - x) \Rightarrow 4 - x = \frac{10}{4} = \frac{5}{2}$

$x = \frac{8-5}{2} = \frac{3}{2}$

$\therefore$ time travelled by aeroplane $= \frac{3}{2}$ hrs.

In 2 hrs, distance travelled by aeroplane = 360 km

When $\frac{3}{2}$ hrs. distance travelled by aeroplane

$= \frac{360}{2} \times \frac{3}{2} = 270$ km

and distance travelled by train

= (360 – 270) = 90 km

23. (a); Let it takes x hrs. in the second case

$\therefore$ Speed $= \frac{1500}{x} = \frac{1500}{x + \frac{1}{2}} + 250$

$\frac{1500\left(x+\frac{1}{2}\right) - 1500x}{x\left(x+\frac{1}{2}\right)} = 250$

$750 = 250x\left(x + \frac{1}{2}\right) \Rightarrow x^2 + \frac{x}{2} - 3 = 0$

$2x^2 + x - 6 = 0 \Rightarrow x = -\frac{4}{2}, \frac{3}{2}; \therefore x = \frac{3}{2}$

$\therefore$ aeroplane takes $\frac{3}{2}$ hrs. in second case.

There in normal case it will take $= \frac{3}{2} + \frac{1}{2} = 2$ hrs.

$\therefore$ speed $= \frac{1500}{2} = 750$ km/hr.

24. (a); Let it takes x hrs in the second case.

$\therefore$ speed $= \frac{120}{x} = \frac{120}{x-1} - 4 \Rightarrow \frac{120}{x-1} - \frac{120}{x} = 4$

$\frac{120(x-x+1)}{x(x-1)} = 4 \Rightarrow 120 = 4x^2 - 4x$

$x^2 - x - 30 = 0 \Rightarrow x = -5, 6; \quad \therefore x = 6$

$\therefore$ the train take 6 hrs. in second case

$\therefore$ the train will take time in normal case

$= (6 - 1) = 5$ hrs.

$\therefore$ original speed $= \frac{120}{5} = 24$ km/hr

25. (d); In 1min. the hare will cover distance

$= 1 \times 60 \times 12 \times \frac{5}{18} = 200$ m

$\therefore$ total distance $= (200 + 100) = 300$ m

relative speed $= (16 - 12) = 4$ km/hr

$= 4 \times \frac{5}{18}$ m/sec. $\Rightarrow$ time to chase $= \frac{300}{4\times5} \times 18$

$= 15 \times 18 = 270$ sec.

In 270 sec. hare cover the distance

$= 270 \times \frac{12\times5}{18} = 900$ metres

Previous Year (Memory Based)

1. (c); Relative speed of Train and man

$= 20$ m/s $- 10$ m/s $= 10$ m/s

Time taken by train to passes the man

$= \frac{180}{10} = 18$ sec.

2. (b); Relative speed of train and man

$= 63 - 3 = 60$ km/hrs

$= 60 \times \frac{5}{18}$ m / sec.

Time $= \frac{500}{60\times5} \times 18 = 30$ sec.

3. (c); Let the length of train is L.

To cross the the platform train travels total distance = 2 L

Speed of train $= 90 \times \frac{5}{18}$ m / s $= 25$ m / s

2L = speed × time

$2L = 25 \times 1 \times 60 \Rightarrow 2L = 1500 \Rightarrow L = 750$ m

4. (c); Let the length of Train = L m.

and the speed of the train is S m/s

$\frac{L+800}{S} = 100$

$L + 800 = 100S \quad$ .... (i)

$L + 400 = 60S \quad$ ... (ii)

Divide eq. (i) by (ii).

$\frac{L+800}{L+400} = \frac{10}{6}$

$6L + 4800 = 10L + 4000$

$4L = 800, \quad L = 200$ m

5. (d); Relative speed of train = 28 km/hr

$28 \times \frac{5}{18}$ m / s

Length of the faster train $= 28 \times \frac{5}{18} \times 18 = 140$ m

6. (b); Let the speed of the second train be x m/s

Speed of first train = 80 km/hr $= 80 \times \frac{5}{18}$ m / s

According to question $= \frac{1000}{x + \frac{80\times5}{18}} = 18$

$1000 = 18x + 400$

$x = \frac{600}{18}$ m / s $= \frac{600}{18} \times \frac{18}{5} = 120$ km / hr

7. (c); Relative speed of trains $= 56 - 29 = 27$ km/h

$= 27 \times \frac{5}{18}$ m / sec.

length of faster train $= 27 \times \frac{5}{18} \times 10 = 75$ m

8. (b); Let the length of each train is L

$S_1 = \frac{L}{3} \qquad S_2 = \frac{L}{4}$

Time taken by trains to pass each other

$= \frac{\text{Total distance}}{\text{Relative speed}}$

Relative speed $= \frac{L}{3} + \frac{L}{4} = \frac{7L}{12}$

$T = \frac{2L}{7L} \times 12 = \frac{24}{7}$ s $= 3\frac{3}{7}$ s

9. **(c);** Let the speed of train is S km/hr and the length of the train is L km.
According to the question

$$\frac{L}{S-3}=10$$

$L = 10S - 30$ ... (i)

$$\frac{L}{S-5}=11$$

$L = 11S - 55$ ... (ii)

Subtract equation 1 from 2.

$$\begin{array}{ccccc} L & = & 10S & - & 30 \\ L & = & 11S & - & 55 \\ - & & - & & + \\ \hline 0 & = & -S & + & 25 \\ S & = & 25 & & \\ \hline \end{array}$$

S = 25 km/hr

10. **(a);** After the Accident, trains speed = 75% of orignal speed.

$$\frac{S_2}{S_1}=\frac{3}{4}$$

Now it takes $\frac{4}{3}$ time of original time.

$$\frac{T_2}{T_1}=\frac{4}{3}\ \left\{S\propto\frac{1}{T}\right\}$$

It train does not stop for an hour then it will be late by 3 hour.

$T_2 = 4 \times 3 = 12$ hr

$T_1 = 3 \times 3 = 9$ hr

Now train takes 12 hr to complete the journey and before it takes 9 hr to complete, the remaining journey.

If accident occur 150 km ahead, time taken by train to complete the remaining journey.

$$\frac{T_2}{T_1}=\frac{4}{3}\times\frac{2.5}{2.5}=\frac{10}{7.5}$$

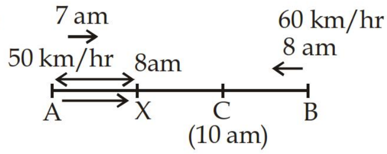
Now it covers the 150 km with original speed in (9 – 7.5) = 1.5

$$=\frac{150}{1.5}=100\text{ km / hr.}$$

Total time of journey = 12 (9 + 3 before Accident)

Total Distance = 100 × 12 = 1200 km.

11. **(b);**



Train from A starts at 7 am with 50 km/hr, in 1 hr it covers 50 km. So at 8 am it is 50 km away from A at X.

These trains meet at C at 10 a.m.

Distance travelled from B to C in 2 hour = 2 × 60 = 120 km.

Distance between X and C = 2 × 50 = 100 km

Distance between AC = 50 + 100 = 150 km

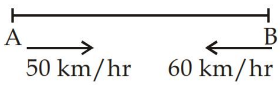
$$\frac{AC}{BC}=\frac{150}{120}$$

AC : BC = 5 : 4

12. **(c);** Relative speed of trains = 5 km/hr
Distance between two trains after 45 min

$$= 5\times\frac{45}{60}=\frac{15}{4}=3\text{ km } 750\text{ m.}$$

13. **(c);**



In 1 hr. second train travels 10 km more than first.

∴ So in 12 hr it travelled 120 km more than the first Relative speed of both trains = 110 km/hr.

Total Distance between AB

= Relative speed × time = 110 × 12 = 1320 km

14. **(c);** Time period for which he runs = $\frac{8000}{250}=32$ sec

Time taken by man to go up = $\frac{24}{10}=2.4$ sec

Time take to go down = $\frac{24}{6}=4$ sec.

Time take to complete a round = 6.4 sec.

Total round = $\frac{32}{6.4}$ = 5 round

15. **(c);** Distance between two trains at 16:30 = 120 km (because aligarh express Runs two hour at a speed of 60 km/hr.)

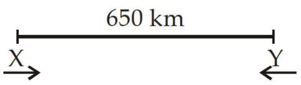
Relative speed = 80 – 60 = 20 km/h

$$\text{Time}=\frac{120}{20}=6\text{ hr.}$$

∴ So, both trains meet 6 hr after 14:30.

Distance between Delhi and meeting point = 6 × 80 = 480 km.

16. (a);



Let the speeds of trains are X and Y km/hr.

$10x + 10y = 650$

$(x + y) = 65$ ... (i)

Distance covered in 8 hours by the trains

$= 65 \times 8 = 520$ km.

$\therefore$ Remaining distance is covered by first train in 4 hours and 20 min.

So speed of first train $= \dfrac{650 - 520}{\left(4 + \dfrac{1}{3}\right)} = 30$ km/hr.

From equation (i) speed of IInd train = 65 – 30 = 35 km/hr.

17. (a); Let the speed by train is S km/hr.

Relative speed of man and train = (S – 4) km/hr.

Total Distance while he see the train

= 50 + 50 + 20 = 120 m

$$\frac{120}{(S-4)\times\frac{5}{18}} = 30, \quad S = 18\frac{2}{5}\text{ km / hr}$$

18. (b); Speed of second train $= 40 \times \dfrac{125}{100} = 50$ km / hr.

Let the speed of third train = S km/hr.

According to question:

$$\frac{20}{S-40} + 1.5 = \frac{25}{S-50}$$

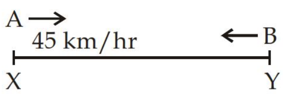
from here we can direct put the value of S from the options.

$\therefore$ S = 60 km/hr.

19. (b); Required Ratio of the speeds of train

$$\frac{S_1}{S_2} = \sqrt{\frac{81}{49}} = 9:7$$

20. (c);



$$\frac{S_A}{S_B} = \sqrt{\frac{200}{288}} = \frac{100}{144} = \frac{10}{12} = \frac{5}{6}$$

$5x = 45$

$6x = 54$

Speed of train B is 54 km/hr.

21. (a); Let the speed of train is S km/hr and the distance = D km

Initially time taken $= \dfrac{D}{}$ hr.

According to question

$$\frac{D}{(S-20)} = \frac{D}{S} + 16 \quad \text{... (i)}$$

$$\frac{D}{(S+10)} = \frac{D}{S} - 5 \quad \text{... (ii)}$$

By solving there two questions

We can get the value of D and S and by using there values we can get time.

By shortcut method:

Distance $= \dfrac{(S_iS_f)(t_2 - t_1)}{(\text{Different in speed})}$

$S_i \rightarrow$ Initial speed

$S_f \rightarrow$ Final speed.

$(t_2 - t_1) \Rightarrow$ Difference in time.

$$\frac{S_i(S_i - 20)}{20} \times 16 = \frac{S_i(S_i + 10)}{10} \times 5$$

$\therefore$ Distance in both cases should be equal.

$8S_i - 160 = 5S_i + 50$

$S_i = 70$ km/hr.

intital speed = 70

Distance $= \dfrac{70 \times 50}{20} \times 16 = 2800$ km.

Actual time $= \dfrac{\text{Dis tan ce}}{\text{speed}} = \dfrac{2800}{70} = 40$ hours.

22. (d); Relative speed of train and man = 55 – 10 = 45 km/hr

Time $= \dfrac{550}{45 \times 5} \times 18$ sec

$= 22 \times 2 = 44$ sec.

23. (c); Relative speed of trains = 36 km/hr.

$= 36 \times \dfrac{5}{18} = 10$ m/s.

length of faster train $= 10 \times 2 = 20$ m.

24. (c); $\dfrac{S_A}{S_B} = \dfrac{3}{4}$

$S \propto \dfrac{1}{T} \Rightarrow \dfrac{T_A}{T_B} = \dfrac{4}{3}$

Let this is 4x and 3x

$4x - 3x = 30$ min

$x = 30$ min

$T_A = 4x = 4 \times 30 = 120$ min = 2 hr.

25. (a); Speed Reduces to $\frac{2}{3}$ the intial speed.

$S_{new} = \frac{2}{3} S$ initial

$\frac{S_n}{S_i} = \frac{2}{3} \Rightarrow \frac{T_n}{T_i} = \frac{3}{2}$

Initial time = 2x.

New time = 3x

$3x - 2x = 1 \quad \Rightarrow \quad x = 1$

$\therefore$ Time to cover the distance with normal speed

$= 2x = 2 \times 1 = 2$ hr.

26. (c); $\frac{S_A}{S_B} = \frac{4}{5} \Rightarrow \frac{T_A}{T_B} = \frac{5}{4}$

Time difference = 15 min

$\frac{T_A}{T_B} = \frac{5 \times 15}{4 \times 15} = \frac{75}{60}$ min.

Distance = $S_A \times T_A = \frac{40 \times 75}{60} = 50$ km.

27. (b); Let the speed of car = x km/hr

and distance of journey = y km.

$y = x \times \frac{9}{2}$ ... (i)

$y = (x + 5) \times 4$ ... (ii)

On dividing (i) by (ii)

$\frac{\frac{9x}{2}}{4x + 20} = 1 \Rightarrow 4x + 20 = \frac{9x}{2}$

x = 40 km/hr

By short cut Method:

Ratio of time. $T_1 \quad T_2$

$\frac{9}{2} \quad 4$

9 : 8

Ratio of speed 8 : 9

9x – 8x = 5

x = 5

Initial speed = 40 km/hr

28. (c); Ratio of speeds of A : B

6 5

Ratio of time = 5 : 6

6x – 5x = 60 + 15

x = 75

Time taken by B = 6x

= 6 × 75 = 450 min = 7 hour 30 min.

29. (c); From the question it is clear that train travels the distance in 11 minutes is equal to the distance covered by sound in 45 second.

Distance travel by train = 330 × 45 m

Time = 11 minutes

Speed = $\frac{330 \times 45}{11 \times 60} \times \frac{18}{5}$ km / hr = 81 km/hr.

30. (d); Total distance travelled by Aeroplane

= 240 × 5 = 1200 km

New speed = $\frac{1200}{5} \times 3$ km / hr = 720 km/hr.

31. (a); Total Distance travelled by car = 40 × 9 = 360 km.

Time taken when this distance travelled by 60 km/hr

$= \frac{360}{60} = 6$ hours.

32. (c); Ratio of speed when he goes by train and walking speed

$\frac{S_T}{S_w} = \frac{25}{4}$

Ratio of time $\frac{T_t}{T_w} = \frac{4}{25}$

4x + 25x = 348 min. [$\therefore$ Total time is given]

29x = 348

x = 12 min.

Time in which he travels by train = 4 × 12

= 48 min

Distance = $25 \times \frac{48}{60} = 20$ km.

33. (a); $\frac{S_1}{S_2} = \frac{3}{2}$

$\frac{T_1}{T_2} = \frac{2}{3}$

2x + 3x = 5, x = 1

he travels 2 hours with speed 3 km/hr.

Total distance = 3 × 2 = 6 km.

34. (b); Let the total distance = 2d.

$\frac{d}{40} + \frac{d}{60} = 10 \Rightarrow \frac{5d}{120} = 10$

d = 240

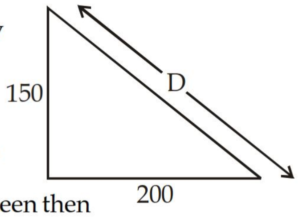
2d = 480 km.

35. (d); Speed of each car in m/s = $36 \times \frac{5}{18} = 10 \text{ m/s}$,

$\Rightarrow 48 \times \frac{5}{18} = \frac{40}{3} \text{ m/s}$

Distance covered by these in 15 sec
= 10 × 15 = 150 m

$\Rightarrow \frac{40}{3} \times 15 = 200 \text{ m}$



Total distance between then

$D = \sqrt{(150)^2 + (200)^2} = 250 \text{ m}$

36. (d); Let the speed of C is S m/s
Speed of B = 3S
Speed of A = 6S
Ratio of their time

$T_A : T_B : T_C = \frac{1}{6} : \frac{1}{3} : \frac{1}{1} = 1 : 2 : 6$

$6x = 72 \Rightarrow x = 12$
A covered the distance in 12 minutes

37. (c); Speed of second train = $\frac{364}{4} = 91 \text{ km/hr}$

According to the question their speeds are 6S : 7S
$7S = 91 \Rightarrow S = 13$
Speed of first train = 6 × 13 = 78 km/hr

38. (c); To jind the ratio of their speeds we need to find the individuals speed in same unit.
So we are going to find the speed in m/s.

Speed of truck = $\frac{550}{60}$

Speed of Bus = $\frac{33 \times 1000}{45 \times 60}$

$\frac{S_{truck}}{S_{Bus}} = \frac{55 \times 45 \times 60}{6 \times 33 \times 1000} = \frac{5 \times 45}{3 \times 100} = \frac{3}{4}$

Ratio = 3 : 4

39. (b); Distance Ratio = 1 : 2 : 3
Time of traveling = 3 : 2 : 1

Speed Ratio = $\frac{\text{Distance}}{\text{Time}} = \frac{1}{3} : \frac{2}{2} : \frac{3}{1} = 1 : 3 : 9$

40. (c); Fixed time = $\frac{20}{5}$ – 40 min = 3h 20 min.

Time taken when he travels 8 km/hr.

= $\frac{20}{8}$ = 2 hours 30 min.

So he reached 50 minutes earlier.

41. (a); Let the speed of Abhay = x km/hr
and the speed of sammer = y km/hr

$$\frac{30}{x} - \frac{30}{y} = 2 \quad \text{... (i)}$$

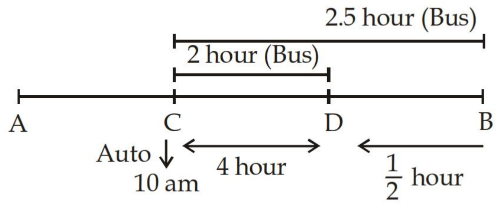
$$\frac{30}{y} - \frac{30}{2x} = 1 \quad \text{... (ii)}$$

On adding (i) and (ii) we get.

$$\frac{30}{x} - \frac{30}{2x} = 3$$

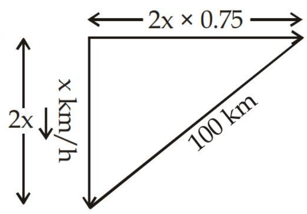
x = 5 km/hr

42. (a);



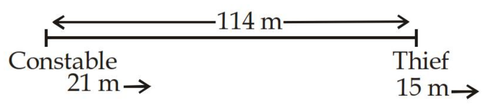
Let C is a point where auto and the bus meet first time and D is a point where the auto and the Bus meets 2nd time.
According to the above diagram;
Distance CD covered by BUS in 2 hour and auto covered the same distance in 4 hours.
Distance CB covered by Bus in 2.5 hr and by Auto in double of Bus.
= 2.5 × 2 = 5 hours
Time at which auto will reach at harid was
= 10 + 5 = 3 P.M.

43. (b);



$(2x)^2 + (1.5x)^2 = (100)^2$
$4x^2 + 2.25x^2 = 10000$
x = 40 km/hr.

44. (a);



∴ Relative speed between Thief and constable
= 21 – 15 = 6 m/min

Time = $\frac{114}{6} = 19$ minutes

45. (c); Let their speeds are 5 and 4 m/s

Relative speed = 1 m/s

Time to caught the thief = $\frac{100}{1} = 100$ sec

Distance travelled by thief = 4 × 100 = 400 m

46. (d); Let initial speed = S km/hr

$$D = \frac{S_i S_f}{\text{Difference in speed}} \times (\text{Difference in time})$$

$$\text{Distance} = \frac{S(S+3)}{3} \times \frac{40}{60}$$

$$\text{2 nd condition } D = \frac{S(S-2)}{2} \times \frac{40}{60}$$

$$\frac{S(S+3)}{3} \times \frac{40}{60} = \frac{S(S-2)}{2} \times \frac{40}{60}$$

(Distance is same in both the cases)

S = 12 km/h

$$\text{Distance} = \frac{12 \times 15}{3} \times \frac{40}{60} = 40 \text{ km.}$$

47. (a); From the question it is clear that Bus stops for a time in which it travels a distance of 10 km with a speed of 50 km/hr.

$$\text{time} = \frac{10}{50} \text{ hr} = \frac{10}{50} \times 60 \text{ min} = 12 \text{ min.}$$

48. (b); Avg speed = $\frac{\text{Total distance}}{\text{Total time taken}}$

$$= \frac{10+12}{\frac{10}{12}+\frac{12}{10}} = \frac{22}{\frac{50+72}{60}}$$

$$= \frac{22 \times 60}{122} = 10.8 \text{ km/hr.}$$

49. (b); Total distance = $30 \times \frac{12}{60} + 45 \times \frac{8}{60} = 12$ km.

Total time = (12 + 8) min = 20 min

$$\text{Aug speed} = \frac{12}{20} \times 60 = 36 \text{ km / hr.}$$

50. (a); Let the total distance is 100 km.

So, 30 km is covered with a speed of 20 km. 1hr.

∴ 60 km at a speed of 40 km 1 hr. and the remaining 10 km distance at 10 km 1hr.

$$\text{Total time} = \frac{30}{20} + \frac{60}{40} + \frac{10}{10}$$

$$= \frac{60+60+40}{40} = \frac{160}{40} = 4 \text{ hr.}$$

$$\text{Avg speed} = \frac{100}{4} = 25 \text{ km/hr.}$$